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Special Collection:

The Energy Exascale Earth
System Model (E3SM):
Advancements in a Decade of
Earth System Modeling

Key Points:

- E3SMv3 introduces major updates to the atmosphere, sea ice, and land components, enhancing capability and improving fidelity
- E3SMv3 increases resolution in the ocean-sea ice mesh and unifies land and river grids for tighter coupling within and between components
- Reduced aerosol forcing in E3SMv3 resolves mid-20th century cooling biases, aligning historical temperature trends with observations

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The Energy Exascale Earth System Model Version 3: 2. Overview of the Coupled System

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Abstract The Energy Exascale Earth System Model version 3 (E3SMv3) represents the latest advancement in Earth system modeling developed by the U.S. Department of Energy (DOE). Building upon previous versions, E3SMv3 introduces significant updates across its coupled components to enhance capability and improve fidelity. The atmosphere component incorporates advancements in chemistry, aerosol-cloud interactions, convection, and microphysics. The ocean features a new time-stepping scheme and a higher-resolution unstructured mesh with sub-ice-shelf cavities, while the sea ice model integrates advanced snow and ice physics for more realistic cryospheric simulations. The land model introduces prognostic vegetation dynamics and a new sub-grid topographic treatment of solar radiation. A new tri-grid configuration harmonizes the horizontal grids of the land and river components for improved process coupling. It is enabled by a new non-linear remapping between the atmosphere and land. E3SMv3 underwent extensive testing through a comprehensive simulation campaign, including pre-industrial control, idealized CO₂ experiments, and historical simulations spanning 1850–2024. The model demonstrates significant improvements in simulating the evolution of the historical surface temperature, particularly addressing the “pothole cooling” bias in earlier versions. Reduced aerosol-related forcing contributes to more realistic radiative forcing and better alignment with the observational record. Ocean heat content (OHC) and sea ice trends are also improved as a result.

Plain Language Summary The Energy Exascale Earth System Model version 3 (E3SMv3), developed by the U.S. Department of Energy, is a state-of-the-science tool designed to advance our understanding of the Earth and energy systems. This model simulates interactions between the atmosphere, land, rivers, oceans, and sea ice to predict Earth system changes and their impacts. E3SMv3 includes major improvements, such as enhanced representations of atmospheric chemistry, clouds, aerosols, sea ice, and

vegetation dynamics. It also introduces refined marine and land grids to improve the accuracy of how these components interact. E3SMv3 was evaluated through simulations of past and historical conditions, totaling over 5,000 years. The model successfully resolved a major issue leading to unrealistic cooling trends during the mid-20th century. E3SMv3 now provides more accurate predictions of surface temperatures, ocean heat content, and sea ice changes, closely matching observed data.

1. Introduction

The Department of Energy (DOE) Energy Exascale Earth System Model (E3SM) project was established a decade ago to develop state-of-the-science Earth system modeling capabilities that optimize the use of DOE resources (Leung et al., 2020). New E3SM major model versions are released every 3–4 years. These releases are scheduled to meet the needs of the E3SM project and DOE and are not aligned with external initiatives such as the Coupled Model Intercomparison Project (CMIP), although the E3SM project participates in such efforts.

Since its inception, E3SM has released two major versions of its physical coupled model, which includes interactive components for the atmosphere, ocean, sea ice, land, and river runoff. E3SM was initially branched from CESM1.2 (Community Earth System Model; Hurrell et al., 2013) but has since evolved significantly, with all components being new or substantially modified. E3SMv1 was released in 2018 (Golaz et al., 2019), followed by E3SMv2 in 2022 (Golaz et al., 2022). These model releases served as the foundation for additional coupled configurations with enhanced capabilities, such as globally higher resolution in all components (E3SMv1.HR; Caldwell et al., 2019), biogeochemistry (E3SMv1.1; Burrows et al., 2020), cryosphere configuration (E3SMv1.2; Comeau et al., 2022), higher resolution around North America (E3SMv2.NARRM; Tang et al., 2023), and improved ocean modeling (E3SMv2.1; K. M. Smith et al., 2025).

The present article documents the third major release of the coupled physical model E3SMv3. Development priorities for E3SMv3 were identified following the initial assessment of E3SMv2 (Golaz et al., 2022) and focused toward improving model performance and enhancing simulation capabilities. The top priority was to improve the model's fidelity in reproducing the observed historical temperature record. This would involve correcting the overly strong aerosol-induced radiative cooling, which caused a “pothole cooling” during 1960–1990 in the temporal evolution of the global mean surface temperature, otherwise expected to show a continual warming trend during the second half of the 20th century. New developments also aimed to resolve deficiencies in simulating both intraseasonal and interannual variability, particularly in the tropics, which carry important signals underlying subseasonal to seasonal predictability. Another important goal was the improvement of the Atlantic Meridional Overturning Circulation (AMOC), which had been persistently weak in previous versions, impacting the model's ability to accurately simulate the redistribution of heat and freshwater around the Earth and its influence on global climate and regional weather patterns.

E3SMv3 approaches these development priorities by augmenting capabilities in individual components and improving the couplings within and between the components through increased resolution and enhanced process representations. The atmosphere incorporates over a dozen new features. The land introduces dynamic vegetation growth and sub-grid topographic effects of solar radiation. The ocean retains new features introduced in the intermediate E3SMv2.1 release (K. M. Smith et al., 2025) and adds a new time-stepping scheme. The sea ice uses the updated physics from the CICE Consortium Icepack v1.5 (Hunke et al., 2024) and advanced morphological features.

E3SMv3 now adopts a tri-grid configuration with three distinct horizontal grids for the atmosphere, ocean, and land. While the atmosphere resolution remains unchanged from v2, the horizontal resolution in the ocean was increased to 30 km or less globally, nearly doubling the grid cell count. To facilitate future water management features, the land and river now share a common grid. As a result, the resolution of the land model increased significantly. A new non-linear remapping algorithm was developed to minimize grid imprinting from the lower resolution atmosphere onto the higher resolution land grid.

These new developments have successfully led to a much improved E3SMv3. Poor representation of interannual and intraseasonal atmospheric tropical variability has been remedied as documented in Part 1 of the E3SMv3

overview series describing the atmosphere component (Xie et al., 2025). E3SMv3 now simulates the warming over the historical observational record more realistically, resolving a major problem that plagued previous versions. The adoption of a mixed-layer submesoscale eddy parameterization has led to an improved AMOC in the intermediate version E3SMv2.1 (K. M. Smith et al., 2025) after the release of E3SMv2. Unfortunately, the gain in the AMOC strength is not maintained after coupling the ocean with other updated components for E3SMv3, suggesting a daunting task of overcoming the weak AMOC with the standard resolution E3SM. Instead, the E3SM project is also finalizing a substantially higher resolution coupled model based on E3SMv3 with an eddy-resolving ocean, which has shown a stronger skill in producing an AMOC comparable to the observations (Caldwell et al., 2019).

This article is Part 2 of the E3SMv3 overview and focuses on the coupled system. It is organized as follows. Section 2 describes changes to the model grids and updates to the components, including coupling changes and computational performance. Section 3 introduces the extensive 5000-year simulation campaign conducted to validate this new model release. Section 4 presents an initial analysis of the simulation campaign, covering the pre-industrial control simulation (Section 4.1), idealized CO₂ simulations for estimating climate sensitivity (Section 4.2), temporal evolution of the historical simulations (Section 4.3), present-day climate including comparison to other CMIP models (Section 4.4) and tropical variability (Section 4.5). Finally, conclusions are discussed in Section 5.

We can only provide an initial overview of E3SMv3 and we choose to focus on key changes from earlier versions, undoubtedly leaving out many interesting aspects. An upcoming Part 3 describing the 25-member historical and near future (2050) large ensemble will cover additional aspects, including variability of the coupled system. Additional topical articles with more in-depth analyses are also planned.

2. Model Description

2.1. Component Grids

The standard resolution configuration of E3SMv3 described in this work adopts a “tri-grid” configuration consisting of three distinct horizontal grids: (a) atmosphere, (b) ocean and sea ice, and (c) land and river. This grid configuration is sometimes referred to below by its E3SM technical name (“*ne30pg2_r05_IcoswISC30E3r5*”). Table 1 summarizes key features of the grid configuration.

The atmosphere uses separate grids for the dynamics and physics as introduced in E3SMv2 (“*ne30pg2*”; Hannah et al., 2021). The dynamics are solved on a cubed-sphere grid with approximate grid spacing of 110 km, whereas the column physics parameterizations are solved on a reduced resolution grid with approximate 165 km spacing. The vertical resolution was increased from 72 to 80 layers by adding 8 layers in the lower stratosphere, improving the representation of stratospheric dynamics while keeping the model top fixed at approximately 60 km (Xie et al., 2025).

In a departure from previous E3SM versions, the land and river components now share a common 0.5° latitude-longitude horizontal grid (“*r05*”). For the land, this corresponds to a very large increase in the number of grid cells from 7,688 in E3SMv2 to 94,838 in E3SMv3, allowing finer representation of spatial heterogeneity in soil, vegetation, and land use. The common grid was introduced both to enhance land resolution and to support more consistent land-river coupling as a foundation for advanced two-way coupling schemes, including irrigation (Yao et al., 2025; Zhou et al., 2020) and inundation (Xu et al., 2022), which have already been applied in other E3SM studies on this grid configuration. These schemes are not activated in the standard configuration presented here, but the new setup establishes the basis for future simulation campaigns that aim to leverage the full suite of land-river coupling improvements.

Note that although the land and river components share this common grid structure, they were historically developed using separate input data sets with slightly different land–sea masks and coastal definitions, resulting in a mismatch in the number of active grid cells—94,838 for land and 99,669 for river, with nearly half of the discrepancy occurring around Antarctica. This mismatch does not affect the overall water budget, as any runoff generated by the land model is still routed to the ocean through the river network, including grid cells where the land model provides no runoff input. Looking ahead, this inconsistency will be addressed through a more tightly integrated data generation framework, in which all components are built from a common foundation with coordinated input data and a consistent land–sea mask.

Table 1
Overview of Component Grids in the Lower Resolution Configurations of E3SMv2 and E3SMv3

Component	E3SMv2			E3SMv3		
	Horizontal		Vertical Layers	Horizontal		Vertical Layers
	Spacing	# Cells		Spacing	# Cells	
Atmosphere (dynamics)	~110 km	48,602	72	Unchanged		80
(physics)	~165 km	21,600	"	Unchanged		"
Land	~165 km	7,688	15 ground 5 snow	0.5°	94,838	Unchanged Unchanged
River	0.5°	99,669	N/A	Unchanged		
Ocean and sea ice	30–60 km	236,853	60 ocean 7 sea ice 5 snow	20–30 km	465,044	64 ocean Unchanged Unchanged

Note. Horizontal grid spacing is reported in kilometer, except for latitude-longitude grids.

The ocean and sea ice components share the same horizontal “IcoswISC30E3r5” mesh illustrated in Figures 1 and 2. The horizontal unstructured mesh was generated using the JIGSAW library (Engwirda, 2017), so that Earth's surface is mapped by an icosahedral mesh, which forms the underlying triangular mesh. The vertices of the triangles are then used as the generators of a spherical Voronoi tessellation, resulting in a companion mesh consisting of hexagons and pentagons. The hexagons have a target resolution of 30 km, but actual cell sizes vary between 21 and 29 km, with most resolution falling near the upper range (Figures 1 and 2). This global mesh is culled to remove any cells that have a land fraction over 0.5 (with a check to preserve key ocean passages), resulting in 465,044 ocean cells, a near doubling compared to the standard E3SMv2 mesh (Table 1). The ocean variables use a C-grid staggering, with the sea surface height (SSH) stored at cell centers, and edge-centered velocities between polygonal vertices on which vorticity is calculated (Ringler et al., 2013). Conversely, the sea ice model uses B-grid staggering with vectors on cell vertices and cell-centered scalars (Turner et al., 2022). Ice-shelf cavities are included in the ocean domain but omitted from the sea ice domain. Coupling on both ocean and sea ice meshes is performed through the centroids, including interpolation of vectors to and from grid centers for outgoing and incoming exchanges, respectively. In the vertical, the number of layers in the ocean increased slightly from 60 to 64.

2.2. Atmosphere

As detailed in Part 1 (Xie et al., 2025), the E3SMv3 atmosphere model (EAM) has undergone substantial developments, particularly in its atmospheric physics, to enhance the model's ability to project future climate changes and reduce outstanding model errors related to aerosols, cloud, and precipitation in E3SMv1 and v2. The vertical resolution has increased from 72 to 80 layers, with extra layers in the lower stratosphere to better represent stratospheric dynamics and the Quasi-Biennial Oscillation (QBO; Yu et al., 2025). Diagnostics have been expanded to include aerosol-cloud interaction (ACI) metrics and the E3SM Chemistry Diagnostics Packages (ChemDyg; H.-H. Lee et al., 2025). Several bug fixes have been implemented, particularly in the convective gustiness scheme (Section 2.6.2 below).

EAMv3 dynamical core is based on EAMv2 with minor updates to the formulation of the pressure gradient term in the momentum equation and the hyperviscosity operator for horizontal dissipation to the prognostic variables. With these changes, EAMv3 is using more realistic topography with less smoothing as compared to EAMv2, leading to better topography simulation and reduced dissipation.

Major atmospheric physics updates include: (a) Updated atmospheric chemistry (Tang et al., 2025) with interactive tropospheric gas chemistry (Prather et al., 2011) and a revised linearized stratospheric ozone scheme (aka Linoz; Hsu & Prather, 2010). (b) Enhanced aerosol representation of stratospheric sulfate (Hu et al., 2025), and secondary organic aerosol (SOA; Lou et al., 2020; Shrivastava et al., 2015). (c) Strengthened coupling of dust mobilization with surface changes based on Kok et al. (2014) and improved linkage between surface emissions and aerosol processes (Vogl et al., 2024; Wan et al., 2024). (d) Improved representation of ice clouds and aerosol-

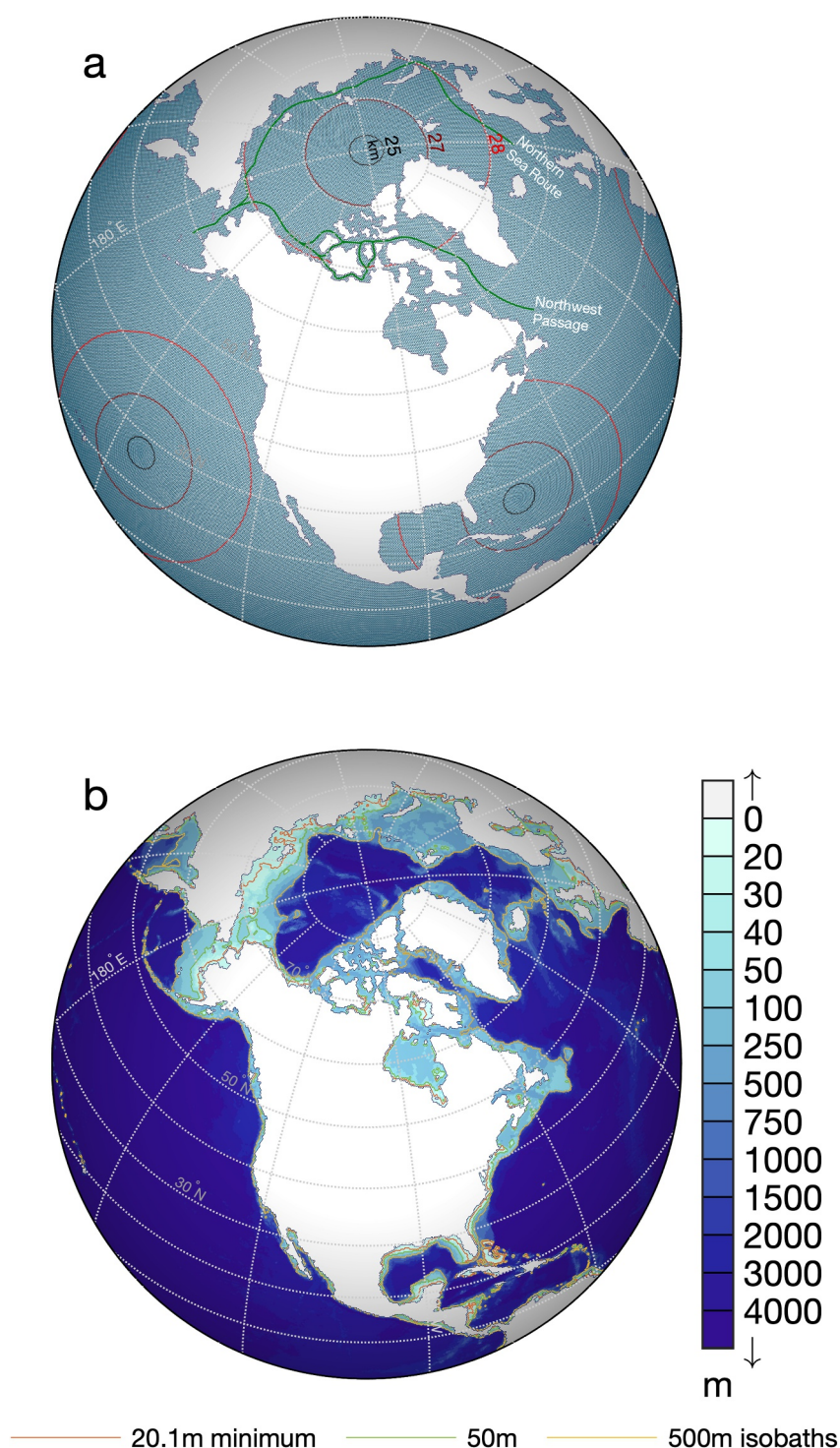


Figure 1. Key North American and Arctic features of the “*IcoswISC30E3r5*” unstructured marine mesh for ocean and sea ice components in E3SMv3: (a) Concentric 25, 27, and 28 km resolution isopleths (reds) represent refinement up to the maximum integer coarseness, which resolves coastal features to eddy-parameterized limits and includes Arctic shipping channels (green); (b) Bathymetry surrounding North America (shaded), including a minimum depth of 20.1 m with two vertical ocean layers up to the approximate draft limit of Arctic landfast sea ice, as well as the 50 and 500 m isobaths. Coastlines are depicted as represented by the unstructured mesh.

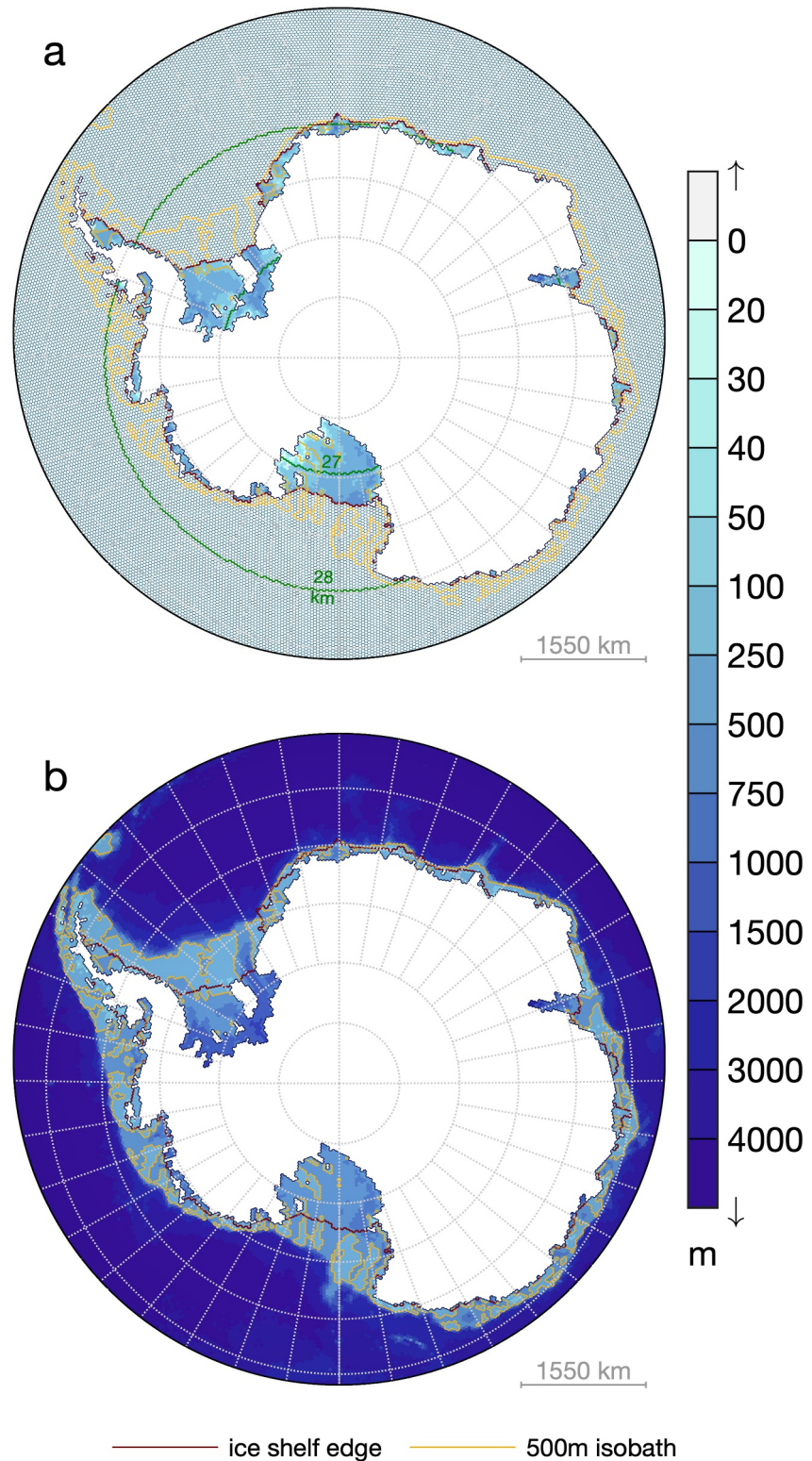


Figure 2. Key Southern Ocean features of the “*IcoswISC30E3r5*” unstructured marine mesh in E3SMv3 illustrating the Antarctic ice shelf edge and 500 m isobath: (a) Antarctic ice shelf cavity water column thickness from the ocean bottom to ice bottom (shaded) with associated polar resolution 27 and 28 km isopleths (green), north of which the resolution is less than 29 km; (b) Associated Southern Ocean bathymetry, including under ice shelves. The colorbar applies to both cavity thickness and ocean depth.

cloud interactions using the Predicted Particle Properties (P3) scheme for stratiform clouds (Morrison & Milbrandt, 2015; Shan et al., 2024; J. Wang et al., 2021). (e) Enhanced Zhang-McFarlane (ZM) deep convection with two-moment cloud microphysics (Song & Zhang, 2011; Song et al., 2025), a new mass-flux adjustment to better connect convection and its environment (Song et al., 2023), and the Multiscale Coherent Structures Parameterization (MCSP, C.-C. Chen et al., 2021; Moncrieff, 2019) for mesoscale organization. The impacts of these new cloud and convection features are discussed in detail in Terai et al. (2025).

Other notable changes include improved physical coupling of aerosol wet removal with convective cloud properties (Shan et al., 2024), retuned dimethyl sulfide (DMS) emission (doubled), reduced monolayers for aging of primary-carbon-mode particles, slightly increased hygroscopicity of particulate organic matter (kPOM), and a minimum in-cloud droplet number concentration (20 cm^{-3}). Collectively, these changes reduce the magnitude of aerosol forcing.

Overall, these updates in physics improve E3SM's ability to simulate climate, tropical variability, and aerosol forcing. The radiation, gravity wave, and the Cloud Layers Unified By Binormals (CLUBB) turbulence and shallow convection schemes remain unchanged.

2.3. Ocean

The ocean component in E3SMv3 uses the same ocean dynamical core as E3SMv2.1 (K. M. Smith et al., 2025). The key differences from v2.1 to v3 are the use of second-order Adams-Bashforth time-stepping for baroclinic momentum advection, which significantly improves the ocean model computational throughput, and the new “*IcoswISC30E3r5*” mesh, which has finer horizontal resolution and includes ice-shelf cavities, which were absent in v2.1.

The ocean component, MPAS-Ocean, solves the hydrostatic, incompressible equations of motion under the Boussinesq approximations. The time-stepping uses a barotropic-baroclinic split: the fast-moving full-depth motions are calculated using a 1-min barotropic time step, while the baroclinic dynamics are calculated using a larger time step of 30 min (or 15 min as discussed below in Section 3). The baroclinic momentum advection has been updated to use second-order Adams-Bashforth time-stepping, while the tracer advection remains Euler forward. The vertical grid uses an arbitrary Lagrangian-Eulerian method, with a z^* coordinate, free surface, and partial bottom cells. The Haney number (a measure of ocean layer slope) is limited to ≤ 10 to limit spurious currents due to pressure gradient errors. Nominal layer thicknesses are 10 m in the upper 200 m, increase gradually from 50 to 200 m between 280 and 2,400 m depth, and reach 240 m for layers deeper than 4,000 m. Importantly, the new v3 ocean mesh geometry includes sub-ice-shelf cavities around Antarctica as illustrated in Figure 2. The minimum water column thickness (e.g., over continental shelves and ice-shelf cavities) is set to 20 m, with a minimum of two ocean layers. The bathymetry is based on GEBCO 2023 (GEBCO Bathymetric Compilation Group 2023, 2023), updated with BedMachine Antarctica v3 (Morlighem, 2022) on the Antarctic shelves.

The ocean parameterizations of mesoscale eddy effects include isoneutral diffusion following Redi (1982) tapered near the surface and slope-limited (with slopes limited to smaller than 0.01), with a $400.0 \text{ m}^2 \text{ s}^{-1}$ constant neutral diffusivity, and a modified Gent and McWilliams (1990) (GM) scheme following Ferrari et al. (2010) using a $900.0 \text{ m}^2 \text{ s}^{-1}$ diffusivity. The mesoscale parameterization coefficients (GM diffusivity and Redi neutral diffusivity) are linearly reduced in regions where the resolution lies between 30 and 20 km, and reach zero for resolutions finer than 20 km. Restratification due to mixed layer eddies is represented by a submesoscale scheme following Fox-Kemper et al. (2011), with a minimum frontal width of 1 km and a 2-day mixing timescale. Ocean surface boundary layer turbulence is represented using the K-Profile Parameterization (KPP) scheme (Large et al., 1994; Van Roekel et al., 2018) as implemented in the CVMix library (<https://github.com/CVMix/CVMix-src>, <https://doi.org/10.5281/zenodo.1000800>) with a critical bulk Richardson number of 0.25. Interior mixing is modeled with a simple shear instability driven scheme (Large et al., 1994). In addition, a constant background vertical viscosity ($10^{-4} \text{ m}^2 \text{ s}^{-1}$) and a horizontal viscosity (using Laplacian and biharmonic coefficients of $1000.0 \text{ m}^2 \text{ s}^{-1}$ and $1.2 \times 10^{11} \text{ m}^4 \text{ s}^{-1}$ respectively) are applied.

2.4. Sea Ice

The sea ice component represents dynamics using the Elastic Viscous Plastic (EVP) discretization with a Wachspress variational B-grid approximation (Turner et al., 2022) on the “*IcoswISC30E3r5*” mesh. Time-

stepping for the dynamics is unchanged from E3SMv2: 120 elastic sub-cycles during each 30-min coupling time step, identical to the internal sea ice dynamics and thermodynamics time steps. The two largest changes to the sea ice component between E3SMv2 and E3SMv3 are in the marine pressure coupling, discussed in the coupling section below, and in the sea ice column package, which we now describe.

E3SM has been converted to use CICE Consortium Icepack v1.5, which includes advanced sea ice morphology. As with previous versions of E3SM, sea ice thickness h is represented by a 5-category distribution, $g(h)$, and level ice melt ponds following Lipscomb (2001) and Hunke et al. (2013), respectively. In addition, snow cover now undergoes metamorphosis as described by Hunke et al. (2024): freshly fallen snow crystals are assigned a diameter and density of 54.526 μm and 100 kg m^{-3} , respectively, aging to a potential maximum dry radius and density of 2800 μm and 400 kg m^{-3} . The aging process includes wind compaction and build-up against ridges. While the prognostic-salinity mushy-layer thermodynamics (Turner & Hunke, 2015) assumes a uniform snow depth over each sea ice thickness category, snow can be compacted and redistributed by wind between level and deformed ice within each thickness category, as well as being blown into open water. This, in turn, impacts the surface energy balance, including by changing albedo calculated by the SNICAR-AD sea ice radiation scheme (Dang et al., 2019). Wind redistribution determines the effective area of bare ice for very small snow depths and the effective area and depth of melt ponds over level ice. Once those areas are determined, the redistributed snow volume over them is known, from which the snow depth for the remaining snow-covered area can be computed and used for radiation balance calculations. Nonlocal wind redistribution of snow is neglected between grid cells in the column physics but the underlying snow distribution is advected with $g(h)$ to conserve mass. As with E3SMv2, snow parameters are constrained by field measurements rather than being manually tuned to obtain a desirable global sea ice distribution.

A variable floe-size distribution was introduced in E3SMv3 as part of the Icepack conversion, but it remains inactive for simulations in this paper for which our ocean surface wave model is switched off. Instead, results shown in this paper apply the Steele (1992) approach to lateral melting, which is the most critical in the marginal ice zone. However, we have updated ice edge simulations in another way: previously a constrained boundary condition $|\mathbf{u}_i| = 0$ was applied to sea ice velocity ($\mathbf{u}_i$) when concentration or specific ice mass fell below 10^{-3} and 10^{-2} kg m^{-2} , respectively. This boundary condition was applied for EVP stability, and differed from the respective concentration and specific cutoffs of 10^{-11} and 10^{-10} kg m^{-2} applied to thermodynamics. But since turbulent flux calculations are themselves dependent on sea ice velocity (following Roberts et al., 2015), low-concentration “ghost ice” accumulated in the marginal ice zone and was subject to less melting than would otherwise occur if sea ice were drifting. This problem was corrected by setting the EVP low concentration and mass cutoffs to the thermodynamic values, without reducing dynamic stability. Over 5,000 simulation years have been performed with this change without numerical instability, causing us to reassess previously established EVP stability criteria.

2.5. Land and River

Several important structural and functional changes were made for the E3SMv3 Land Model (ELMv3) compared to ELMv2. As mentioned above, the number of land grid points increased by a factor of ~ 12 from ELMv2 to ELMv3. This increase in resolution was associated with the land grid shifting from one that follows the atmosphere to a shared grid between ELM and the river model (MOSART). As with ELMv2, ELMv3 uses a hierarchically nested subgrid scheme that represents land surface heterogeneity within each grid cell (as illustrated in Figure 2 of Lawrence et al., 2019). Variability in geomorphology, soil and snow column physics, and vegetation community are captured in order from coarser to finer nested levels of subgrid variability. At the vegetation community level, the vegetation canopy is treated as dynamic sunlit and shaded fractions as a final level in the subgrid scheme. With higher spatial resolution at the grid level in ELMv3, some of the variability represented at subgrid levels in ELMv2 is now represented as spatially explicit grid variability. With nominal grid cell size of 50 $\text{km} \times 50 \text{ km}$ (at the equator), there is still considerable land surface variation represented in the hierarchical subgrid scheme in ELMv3.

ELMv3 includes two significant functional advances compared to ELMv2. While the v2 coupled model simulations used constant data sets to define the amount of leaf area in the vegetation canopy and other physical attributes of the vegetation structure, the v3 model uses a prognostic model of carbon and nutrient dynamics in plants and soils to update the vegetation structure and canopy leaf area (Thornton et al., 2009; X. Yang

et al., 2019, 2023). This prognostic land ecosystem model updates on the main land time step (30 min), and the prognostic vegetation state is coupled dynamically to energy and water fluxes at the land surface and in the subsurface. The prognostic land ecosystem sub-model in ELMv3 was used previously to explore carbon and nutrient cycle feedbacks with the climate system in the E3SMv1.1-BGC model (Burrows et al., 2020). A benefit of moving from the constrained canopy data sets of ELMv2 to the prognostic vegetation structure of ELMv3 is that the land surface can respond more realistically to changes in atmospheric conditions (e.g., temperature, humidity, precipitation, sunlight, and carbon dioxide concentration), and those responses can be realized seamlessly across historical and future scenarios. Where ELMv2 had a single annual cycle of leaf area for each vegetation type in each gridcell that was repeated for every model year, ELMv3 vegetation structure responds on seasonal, interannual, and longer timescales to variations in the atmosphere and associated variations in land water and energy budgets.

The second functional advance included in ELMv3 is a representation of the effects of subgrid topographic variation on solar radiation absorbed by and reflected from the land surface (Hao et al., 2021). The surface radiation scheme used in ELMv2 assumed that the land surface was a flat and level plane for each grid cell. The ELMv3 scheme takes into account the subgrid terrain details to estimate the influence of angled terrain, shadowing, and multiple scattering paths between adjacent terrain elements. The new sub-model represents variability in these terrain factors under different illumination angles and different proportions of direct and diffuse radiation. The sub-model modifies the land surface albedo without altering the incoming atmospheric radiation.

The river component in E3SMv3 (MOSARTv3) continues to route runoff produced by ELM to the river mouth as freshwater input to the ocean component, with no significant changes to its physics or resolution compared to the previous version. MOSARTv3 retains the kinematic wave approach to route streamflow across hillslopes, tributaries, and main river stems on an eight-direction-based river network (Li et al., 2013) at a 0.5° latitude-longitude spatial resolution. One notable addition in MOSARTv3 is a new water budget tracking module that provides detailed diagnostics of river water fluxes into and out of the coupler at daily, monthly, and annual timescales. This feature enhances the ability to monitor and evaluate water budget consistency across model components.

2.6. Coupler

As in E3SMv2 and E3SMv1, the driver and coupler for E3SMv3 is cpl7 (Craig et al., 2012). The driver of cpl7 performs the integration of the coupled model and provides the “main” for the single executable. cpl7 relies heavily on the Model Coupling Toolkit (Larson et al., 2005) for data types and methods to perform inter-component communication and remapping operations. We refer to this software as “cpl7/MCT.”

Like other model couplers, cpl7/MCT remaps fields from one component's mesh to another. In E3SMv3, cpl7/MCT still reads remapping weights from files prepared ahead of time. These weights are now all computed by TempestRemap (Ullrich & Taylor, 2015; Ullrich et al., 2016) including for bilinear maps which were previously computed using ESMF. This choice was made due to TempestRemap's native support for spectral element atmospheric grids which simplified our model configuration toolchain.

E3SMv3 introduces some new features in cpl7/MCT described in the sub-sections below. A new non-linear map option provides smoother mapped fields while still providing global conservation. There was also a refactoring and correction of the use of gustiness in the momentum flux calculation as well as improved surface pressure coupling.

2.6.1. Nonlinear Maps

At the equator, the half-degree land grid has approximately three times the resolution as the atmosphere model's physics grid; toward the poles, the difference is even greater. Since there is no horizontal mixing in the land surface, conventional area-averaged maps for fluxed quantities from the atmosphere to the land imprint the atmosphere physics grid onto the land fields. In E3SMv3, we introduced an option to map fluxes using a high-order map, after which a global fixer is applied to restore mass conservation and bounds preservation. The high-order map produces much smoother fields on the land grid, while the fixer provides the conservation and shape preservation inherent to the conventional area-averaged map. We refer to the method as a *nonlinear map* because the property-preservation step is nonlinear.

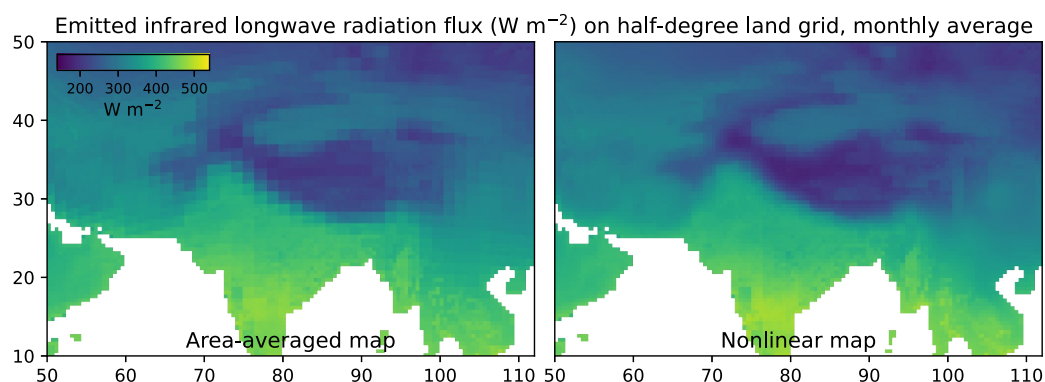


Figure 3. Comparison of the effects of using conventional area-averaged maps (left) and new nonlinear maps (right) for atmosphere-to-surface fluxes. The land model's emitted infrared longwave radiation flux (W m^{-2}) is shown on the land grid. Color limits are the same in both plots.

The nonlinear map has two main ingredients: first-order and second-order finite-volume maps computed by TempestRemap; and an implementation of the CLIPANDASSURED SUM method, a global fixer described in Bradley et al. (2019), specialized to the setting provided by cpl7/MCT. Details are provided in Appendix A. In the E3SMv3 standard-resolution model, nonlinear maps are used to map most fluxes from the atmosphere component to the land, river runoff, ocean, and sea ice components. However, precipitation fluxes are mapped with conventional linear maps. Details concerning this choice are also in Appendix A.

Figure 3 compares the results of the nonlinear map with those of the conventional area-averaged map in the atmosphere-to-land case. Data for the figure were generated using two pre-industrial control simulations, each run for 2 months: one with the default v3 configuration and another with the nonlinear maps replaced by area-averaged maps. The monthly average of the land model's emitted infrared longwave radiation flux, from month two, is plotted on the land grid. To show the atmosphere grid's imprint on the field in the case of the area-averaged map, we zoom to one part of the globe, centered roughly on the Indian subcontinent. The latitude-longitude projection of the atmosphere spectral-element grid is clearly imprinted on the field when using the area-averaged map. The nonlinear map substantially decreases this imprint.

2.6.2. Atmospheric Gustiness

EAM contains a sub-grid parameterization of gustiness (Ma et al., 2022; Zeng et al., 2002). Previously, the coupler used a single total wind measure rather than separate mean velocity and gustiness fields, preventing accurate surface stress calculations. In E3SMv3, this was corrected so that momentum fluxes at the atmosphere-ocean and atmosphere-land interfaces now approach zero as the mean relative velocity does. Gustiness calculations in EAM were also corrected to eliminate an error and double counting over land, affecting sensible, latent, and momentum fluxes but not tracer fluxes, since gustiness effects are not applied to tracers. Testing of these changes showed they had little net effect on the simulated climate (not shown).

2.6.3. Surface Pressure Coupling

A major change was introduced to air-sea pressure coupling in E3SMv3. In previous versions of the model, MPAS-Ocean was coupled to EAM using mid-level surface layer atmospheric pressure. This was instituted to alleviate a sea ice-ocean instability from super-inertial ice-ocean-atmosphere momentum transfer in the presence of sea ice pressure on the ocean surface. Including non-levitating sea ice in a coupled model was illustrated to be important by Hibler et al. (2006), but when introduced into E3SMv1, it caused unusual oscillations in ocean and sea ice velocity in the Canadian Archipelago. These oscillations were thought to be the result of combined sea level and sea ice pressure acting on shallow water. In fact, the oscillations were the result of a coding error in the implementation of wind drag relative to ice velocity in E3SMv1; that error has been corrected and full mean sea level pressure now acts on the ocean surface. The change has greatly reduced ocean and sea ice biases as discussed in Section 4.1.

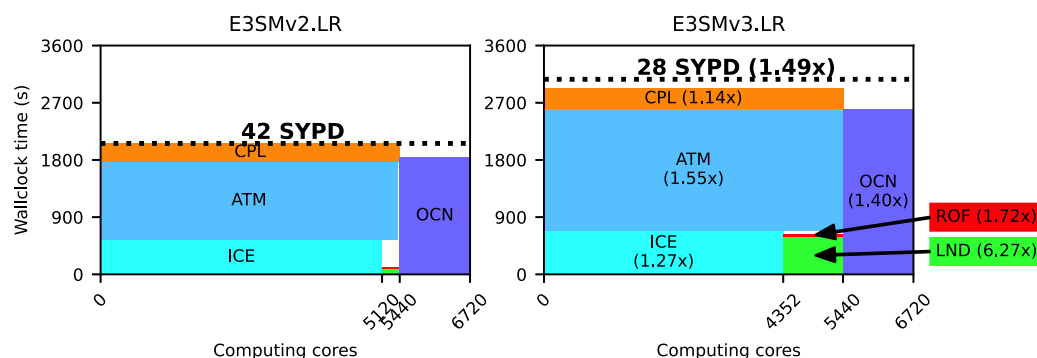


Figure 4. Time-resource product plots illustrating the performance of E3SMv2 and E3SMv3 *piControl* low resolution simulations. Each component is represented as one rectangle. A rectangle has the area given by the product of wallclock time for one simulated year and number of cores. The horizontal dashed lines indicate overall throughput in simulated years per wallclock day (SYPD). E3SMv3 computational cost relative to v2 are indicated in parentheses on the right panel.

2.7. Tuning

The coupled tuning objectives for the pre-industrial control simulation were similar to those used in previous versions (Golaz et al., 2019, 2022). The overall goals were to achieve:

1. Near-zero long-term mean net top-of-atmosphere (TOA) energy flux and total ocean heat content (OHC) in equilibrium.
2. Minimal long-term drift in global mean surface air temperature.
3. A reasonable absolute value of the global mean surface air temperature.

In addition, spatial root-mean-square errors (RMSEs) relative to observations were evaluated for key climate variables, including annual-mean sea surface temperature (SST), annual and seasonal precipitation, TOA radiation, cloud radiative effect, and sea surface wind stress. Most tuning was carried out at the component level, particularly in the atmosphere model during the development of EAMv3 (Shan et al., 2024; Terai et al., 2025), with a focus on the energy budget and aerosol effective radiative forcing. The resulting atmospheric tunings were found to perform well in coupled mode. Dust burden, Antarctic ozone hole characteristics, and column ozone were also closely monitored and adjusted during both uncoupled and coupled stages, in response to the adoption of the new dust scheme, the interactive gas chemistry, and the additional layers in the lower stratosphere. Additional aerosol tuning was performed later in the coupled integration to improve the partitioning of aerosol radiative forcing. No further tuning was applied after the coupled configuration was transitioned to the new ocean mesh and tri-grid.

2.8. Performance

Both E3SMv2 and v3 pre-industrial control simulations were run using 105 nodes on the same Argonne National Laboratory Chrysalis cluster. Chrysalis has 512 compute nodes. Each node has two AMD Epyc 7532 “Rome” 2.4 GHz processors, and each processor has 32 cores, for a total of 64 cores per node. Each node has 256 GB 16 channel DDR4 3,200 MHz memory. The inter-connect hardware is Mellanox HDR200 InfiniBand and uses the fat tree topology. Overall computational cost of E3SMv3 increased by 49% relative to v2, with the throughput decreasing from 42 simulated years per wallclock day (SYPD) in v2 to 28 SYPD in v3.

Figure 4 illustrates efficiency differences by plotting the wall-clock-time-resource product for each component as a rectangle for the 105-node layout for one simulated year. The dashed line at the top shows the overall throughput; because the time value of each component is approximate, the filled rectangles do not sum exactly to the total time value. The atmosphere (ATM), sea ice (ICE), coupler (CPL), land (LND), and river runoff (ROF) components run on one set of nodes, while the ocean (OCN) component runs concurrently on another set. The largest contributors to the increased computational cost from E3SMv2 to v3 are the atmosphere (1.55x), the ocean (1.40x), sea ice (1.27x) and the coupler (1.14x). The cost of the land component increases substantially due to its large increase in horizontal cells (6.27x), but its relative cost compared to the entire model remains small. To

Table 2
Summary of E3SMv3 Simulations

Label	Description	Ocean dt	Period	Ens.	Initialization
Fully coupled (atmosphere, ocean, sea ice, land, and river)					
<i>piControl</i>	Pre-industrial control	30 min	500 years	–	Pre-industrial spinup
<i>piControl_bonus</i>	Rerun of years 401–500 of <i>piControl</i> with extra output for ML training				
<i>1pctCO2_bcdt15m</i>	Prescribed 1% yr ^{−1} CO ₂ increase	15 min	150 years	1	<i>piControl</i> (101)
<i>abrupt-4xCO2_bcdt15m</i>	Abrupt CO ₂ quadrupling	15 min	150 years	1	<i>piControl</i> (101)
<i>historical_N</i>	Historical	30 min	1850–2024	5	<i>piControl</i> (51, 101, 151, 201, 251)
<i>historical_0101_bcdt15m</i>	Historical	15 min	1850–2024	1	<i>piControl</i> (101)
<i>hist-GHG_N</i>	DAMIP well-mixed greenhouse-gas-only historical	30 min	1850–2024	3	<i>piControl</i> (101, 151, 201)
<i>hist-aer_N</i>	DAMIP anthropogenic-aerosol-related historical	30 min	1850–2024	3	<i>piControl</i> (101, 151, 201)
<i>hist-all-xGHG-xaer_N</i>	Other forcing historical (all forcings except GHG and aer)	30 min	1850–2024	3	<i>piControl</i> (101, 151, 201)
Prescribed SST and sea ice extent (atmosphere, thermodynamic sea ice, land and river)					
<i>amip_N</i>	Atmosphere with prescribed SSTs and sea ice concentration	–	1870–2014	3	<i>historical_N</i> (1870)
<i>piClim-control</i>	RFMIP baseline control	–	50 years	–	<i>piControl</i> (51)
<i>piClim-histall_N</i>	RFMIP time-varying ERF all agents	–	1850–2024	3	<i>piControl</i> (101, 151, 201)
<i>piClim-histaer_N</i>	RFMIP time-varying ERF aerosols	–	1850–2024	3	<i>piControl</i> (101, 151, 201)

Note. Within ensembles, *N* is set to the initialization year to distinguish each realization.

accommodate the increase in land cost while maintaining overall computational balance, the land in E3SMv3 runs on a larger set of computational cores.

3. Simulation Campaign

Table 2 summarizes the E3SMv3 simulation campaign that covers the Coupled Model Intercomparison Project Phase 6 (CMIP6) Diagnosis, Evaluation, and Characterization of Klima (DECK) plus historical simulations (Eyring et al., 2016). All simulations were configured according to the CMIP6 specifications and rely on the same boundary files as E3SMv1 (Golaz et al., 2019) and E3SMv2 (Golaz et al., 2022) with a few exceptions. In the latest campaign, (a) new input data streams are required for the tracer variables introduced to the E3SMv3 atmosphere model to enable interactive chemistry (Tang et al., 2025; Xie et al., 2025); (b) volcanic forcing is represented by volcanic sulfur dioxide (SO₂) emissions spanning troposphere and stratosphere—rather than stratospheric sulfate aerosol burden alone for shortwave scattering and longwave absorption—to work with MAM5-PSA and represent not only the direct radiative effect but also the ensuing volcanic aerosol-cloud interaction (Ke et al., 2025); and (c) new input data streams are also required by the dynamic vegetation growth model adopted by the ELMv3 in place of prescribing leaf and stem area indices.

Simulations in Table 2 do not include thermally active Antarctic ice shelves in the manner first demonstrated in E3SM by Comeau et al. (2022); there is no freshwater exchange at the shelf-ocean interface. Rather, ice shelf cavities in Figure 2 were included in simulations listed in Table 2 because they expand the ocean domain to a realistic footprint more than 1.5×10^6 km² larger than if the cavities were neglected. Including Southern Ocean cavities impacts the barotropic mode in MPAS-Ocean as far north as the Labrador Sea, illustrating their global importance regardless of local melt rates (Pal et al., 2023). E3SMv3 simulations that add these melt fluxes with associated changes to ocean physics (including vertical turbulent mixing) are the subject of a separate article.

We initialized the ocean component by interpolating fields of potential temperature and practical salinity from the World Ocean Atlas 2023 (WOA23; Reagan et al., 2024) onto the model's *IcoswISC30E3r5* mesh. The ocean was initially at rest. To suppress fast surface-gravity-wave noise arising from the density structure adjustment, we carried out a 100-day dynamic adjustment using the standalone ocean model forced only by a constant surface wind stress field derived from the National Centers for Environmental Prediction (NCEP) 1958–2000 climatology (Kalnay et al., 1996). During the dynamic adjustment we employed exponential (Rayleigh) drag on the

ocean velocity field: first 10 days at a drag coefficient of $1 \times 10^{-4} \text{ s}^{-1}$, then 10 days at $1 \times 10^{-5} \text{ s}^{-1}$, followed by 80 days with no Rayleigh drag. Once the surface-wave adjustment is complete, we initialized the sea ice component with a uniform 1-m thick ice disc poleward of 70°N and 60°S , and then ran a coupled ocean–sea ice simulation for one month with atmospheric and land forcings taken from the JRA-55 reanalysis (Japanese 55-year Reanalysis; Kobayashi et al., 2015). The ocean and sea ice restart files produced at the end of this month were then used to initialize the fully coupled spin-up.

The land component with prognostic vegetation structure was initialized using a two-step procedure to ensure equilibrium of soil carbon and nutrient pools. Driven by atmospheric forcings from the coupled simulation of a late stage development version of E3SMv3 without dynamic vegetation growth, the model was first run 250 years with accelerated soil decomposition, following the approach of Koven et al. (2013), then another 250 years using the standard decomposition rates. The total leaf area index stabilized after the cumulative 500 years. The restart file at the end was used to initialize the coupled spin-up.

The coupled system was then spun-up for 2000 years under pre-industrial conditions to achieve a quasi-equilibrium as discussed in Section 4.1 below. The official pre-industrial control (*piControl*, 500 years) was a continuation from this state. The last 100 years of *piControl* were rerun a second time with additional high frequency output in the atmosphere, ocean and sea ice components to generate a training data set for Machine Learning (ML) emulators (see, e.g., Duncan et al., 2024). The idealized CO_2 simulations (*1pctCO2* and *abrupt-4xCO2*, 150 years each) and a five-member ensemble of historical simulations (*historical_N*; 1850–2024) were initialized from *piControl* on January 1 of various years as indicated in Table 2. The historical simulations were extended to 2024 using SSP2-45 forcings after 2014 (O'Neill et al., 2016) in an attempt to gauge the model's skill in simulating historical global mean surface temperature up to the latest observational records. Studies (Jones et al., 2021; Y. Yang et al., 2022) have shown that even the strong emissions reductions related to the COVID pandemic in 2020 had a very minor change on the global mean surface temperature, compared to the SSP2-45 baseline, although some discernible effects on regional temperature and rainfall were detected in models. The bulk of the analysis presented in this overview still focuses on the standard CMIP6 historical period through 2014.

The larger baroclinic time step in the ocean is 30 min for the spin-up, pre-industrial control, and all five historical ensemble members. However, the 30 min baroclinic time step was found to be unstable for stronger idealized CO_2 forcing, namely the 1% and $4x\text{CO}_2$ experiments. While the grid spacing is nominally 30 km, actual grid cell sizes vary. Numerical instabilities developed near the smallest cells of 20 km diameter and under strong currents. These instabilities manifested themselves in the thickness-velocity field. We inferred that the baroclinic time step of 30 min is close to the limit of stability for the smallest cells in the mesh (a rule of thumb is that the time step should generally be at most 1 min per km horizontal resolution). Thus, we reduced the baroclinic time step to 15 min for both 1% and $4x\text{CO}_2$ experiments (indicated with the suffix *_bcdt15m*). We verified that this time step change does not have a significant impact on climate with an additional historical simulation *historical_0101_bcdt15m*. We examined a number of global fields and metrics across atmosphere and ocean components (including trend in OHC) and found that this additional historical simulation did not significantly differ from the ensemble run with a 30 min time step. Some of these metrics are shown in Figures 18 and 24, discussed below, where they can be compared with the historical ensemble members using the default 30 min time step. We also find that the Equilibrium Climate Sensitivity (ECS) is not significantly different among stable $4x\text{CO}_2$ simulations with baroclinic time step from 10 to 20 min (not shown).

AMIP (Atmospheric Model Intercomparison Project; Gates, 1992) type simulations (prescribed SST and sea ice extent) were also performed to cover the entire period for which CMIP6 provides surface boundary conditions (1870–2014). Atmosphere, land, and river initial conditions for *amip_N* were taken from year 1870 of the corresponding *historical_N* coupled simulations.

To understand the relative importance of different forcing agents, a set of DAMIP (Detection and Attribution Model Intercomparison Project; Gillett et al., 2016) type historical simulations was performed. They consist of three-member ensembles with well-mixed greenhouse-gases-only (*hist-GHG*) and anthropogenic-aerosol-related (*hist-aer*) forcing agents. The forcing agents in the *hist-GHG* simulations also include chemical precursors that influence the production of methane, nitrous oxide, and tropospheric and stratospheric ozone through the interactive chemistry and Linoz (Hsu & Prather, 2010). Instead of natural-only historical simulations as in Gillett et al. (2016), we opted for a third set with all agents active except well-mixed GHGs, aerosols and aerosol

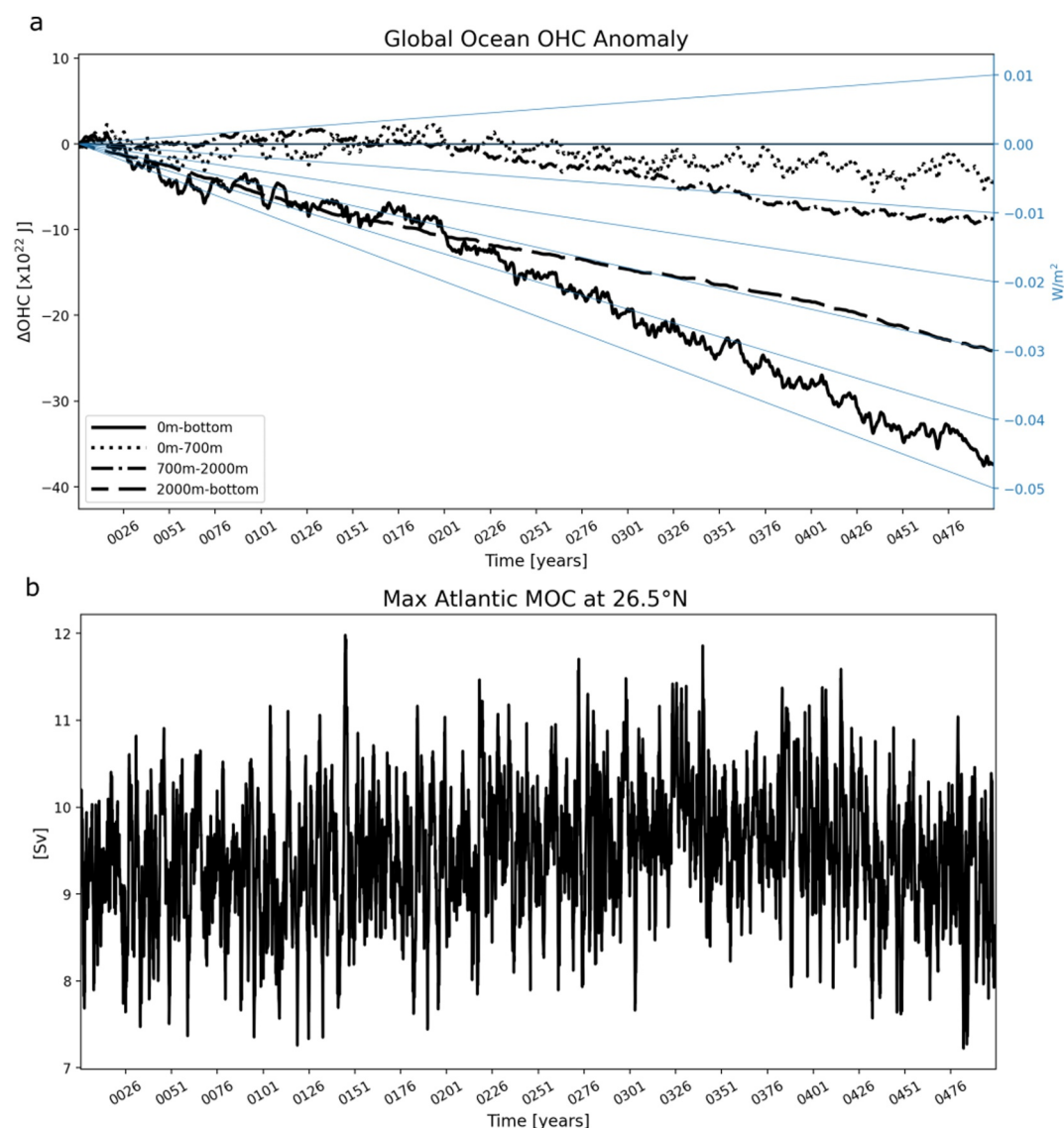


Figure 5. (a) Global mean OHC anomalies over the full 500-year *piControl* simulation over a range of depths. Left axis shows actual change in ocean heat and right axis with blue slanted lines shows the change converted to equivalent energy flux at the top-of-atmosphere. (b) Time series of maximum AMOC at 26.5°N. RAPID Observations have a mean strength of 17 Sv and a standard deviation of 2.8 Sv (McCarthy et al., 2025).

precursors (*hist-all-xGHG-xaer*). This non-standard choice was motivated by a desire to include all forcing agents in our decomposition (including transient land-use and nitrogen deposition and population density that modulate vegetation growth).

We also perform a set of RFMIP (Radiative Forcing Model Intercomparison Project; Pincus et al., 2016; C. J. Smith et al., 2020) simulations to quantify the historical effective radiative forcing (ERF) from both all forcing agents combined and aerosols specifically. These simulations use fixed sea surface temperature and sea ice cover at pre-industrial levels (derived from the *piControl* simulation), which allows isolation of the radiative impact of forcing agents without the feedbacks from the ocean and ice components. The simulations include: (a) *piClim-control*, the baseline simulation with pre-industrial forcing conditions; (b) *piClim-histall*, which includes transient historical changes in all forcing agents (greenhouse gases, aerosols, land use, etc.); and (c) *piClim-histaer*, which includes the historical changes of anthropogenic aerosols and their precursors while keeping other forcing agents at pre-industrial levels. By comparing these simulations, we can quantify both the total historical radiative

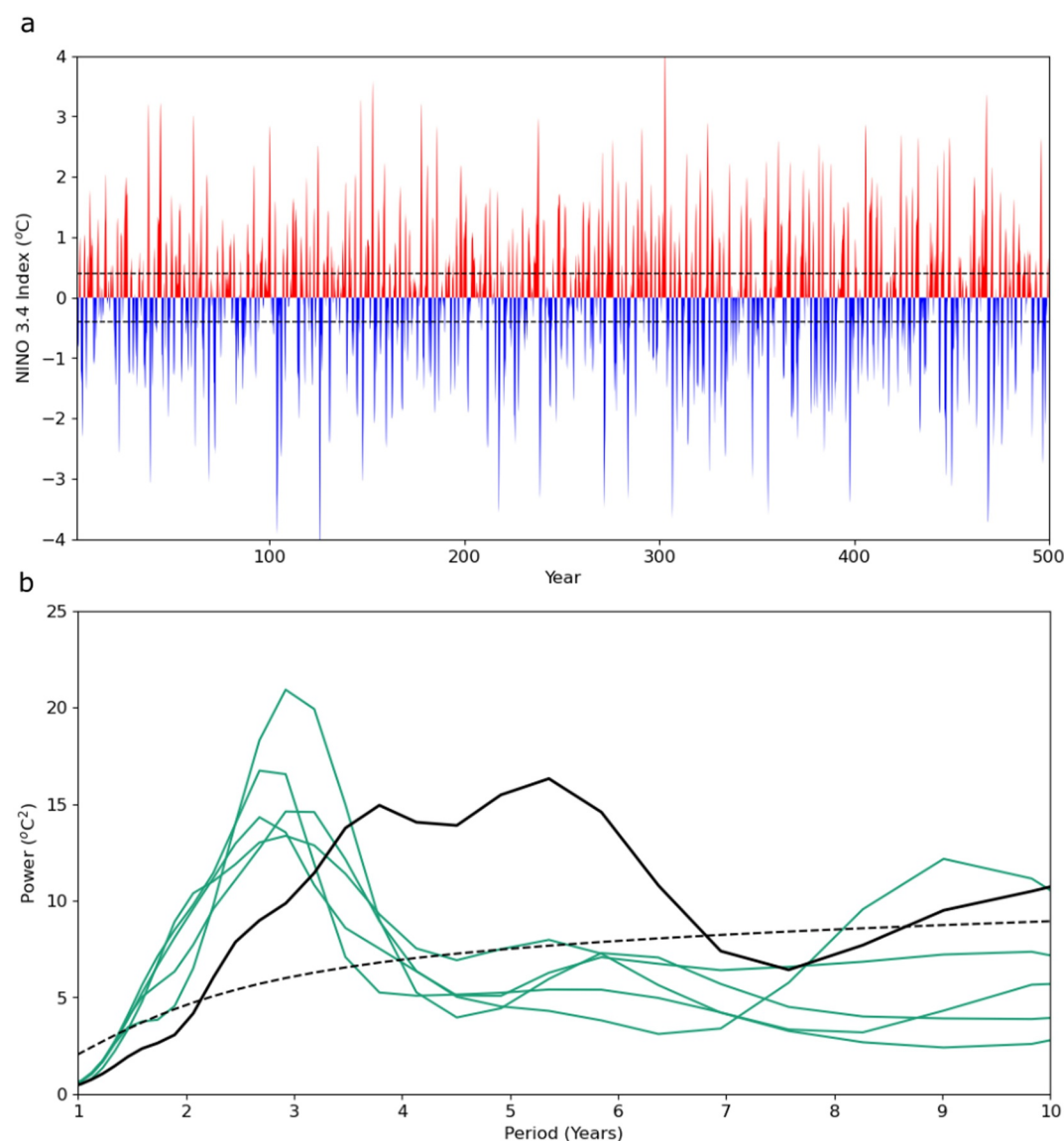


Figure 6. (a) Nino3.4 index time series over the 500-year *piControl* simulation. Blue anomalies below the lower dashed horizontal line are La Niña events and red anomalies above the upper dashed horizontal line are El Niño events. (b) Power spectrum analysis of the Nino3.4 index time series. The *piControl* simulation (green lines) has been divided into five 100-year segments. ERSSTv5 data from Huang et al. (2017) is shown as the thick black line. The 90% confidence interval is shown as the dashed black line.

forcing and the specific contribution from aerosols. The resulting ERF estimates provide important context for understanding climate responses in our historical simulations, as discussed in Section 4.3.

4. Analysis of Simulation Campaign

4.1. Pre-Industrial Control Simulation

The 500-year pre-industrial control simulation (*piControl*) was initialized following a 2000-year long spin-up simulation. The model was identical for both simulations and did not experience any numerical failure during those 2500 simulated years.

TOA net radiation during the 500-year *piControl* averaged -0.05 W m^{-2} with no discernible long-term trend (not shown). Compared to anthropogenic forcing, this overall energy imbalance is sufficiently close to zero for a fully

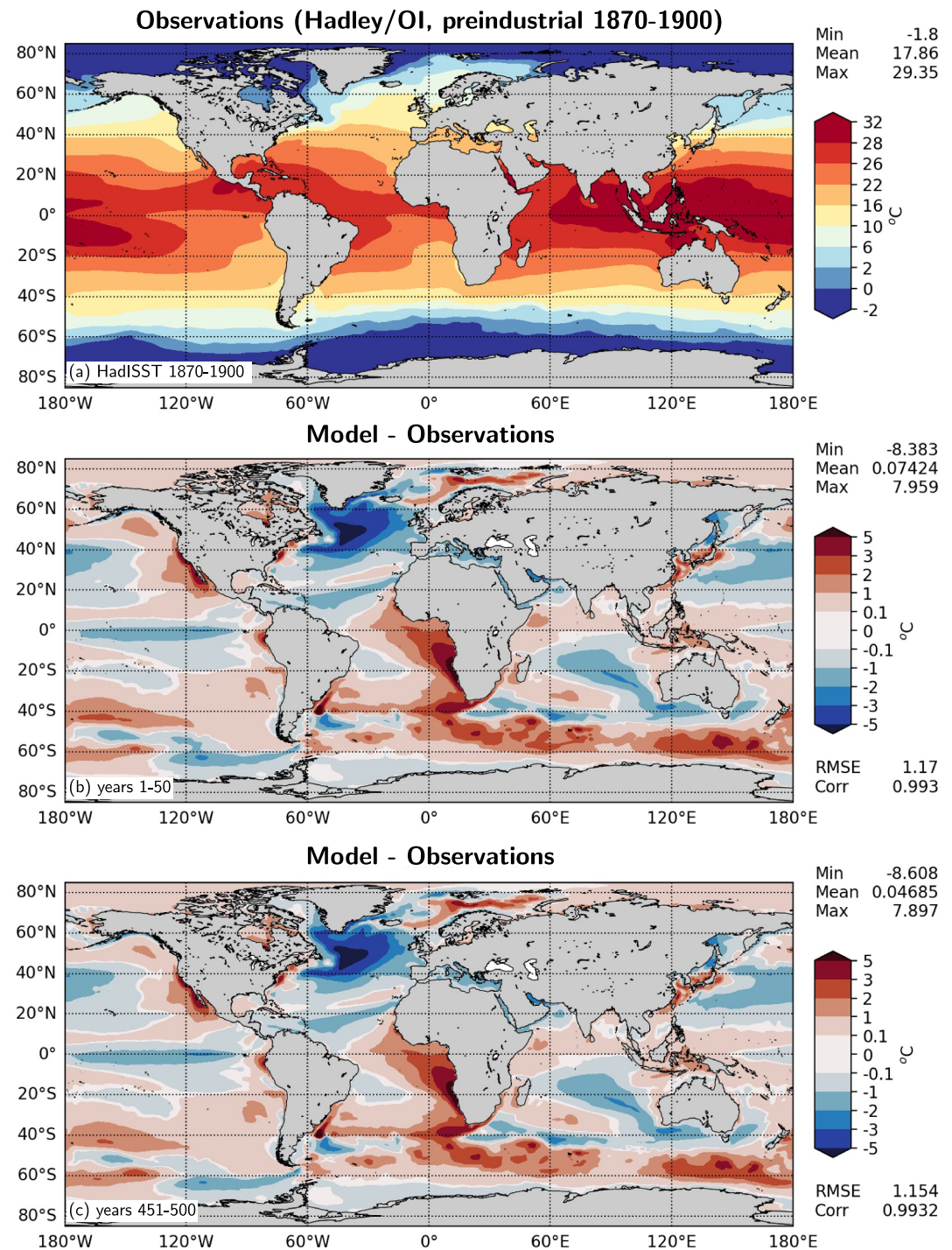


Figure 7. Pre-industrial annual mean sea surface temperature from (a) HadISST (1870–1900; Hurrell et al., 2008); and E3SMv3 *piControl* simulation biases for (b) the first 50 years (1–50) and (c) last 50 years (451–500).

equilibrated and energy conserving model. This small TOA imbalance is consistent with long term evolution of total OHC as depicted in Figure 5a. Total ocean heat loss averages just under 0.05 W m^{-2} (solid line) with most of the contribution (0.03 W m^{-2}) attributable to ocean heat loss to bottom waters (below 2,000 m; dashed line). The upper 700 m water masses (dotted line) experience a net heat loss an order of magnitude smaller at 0.005 W m^{-2} when averaged over the simulated 500 years.

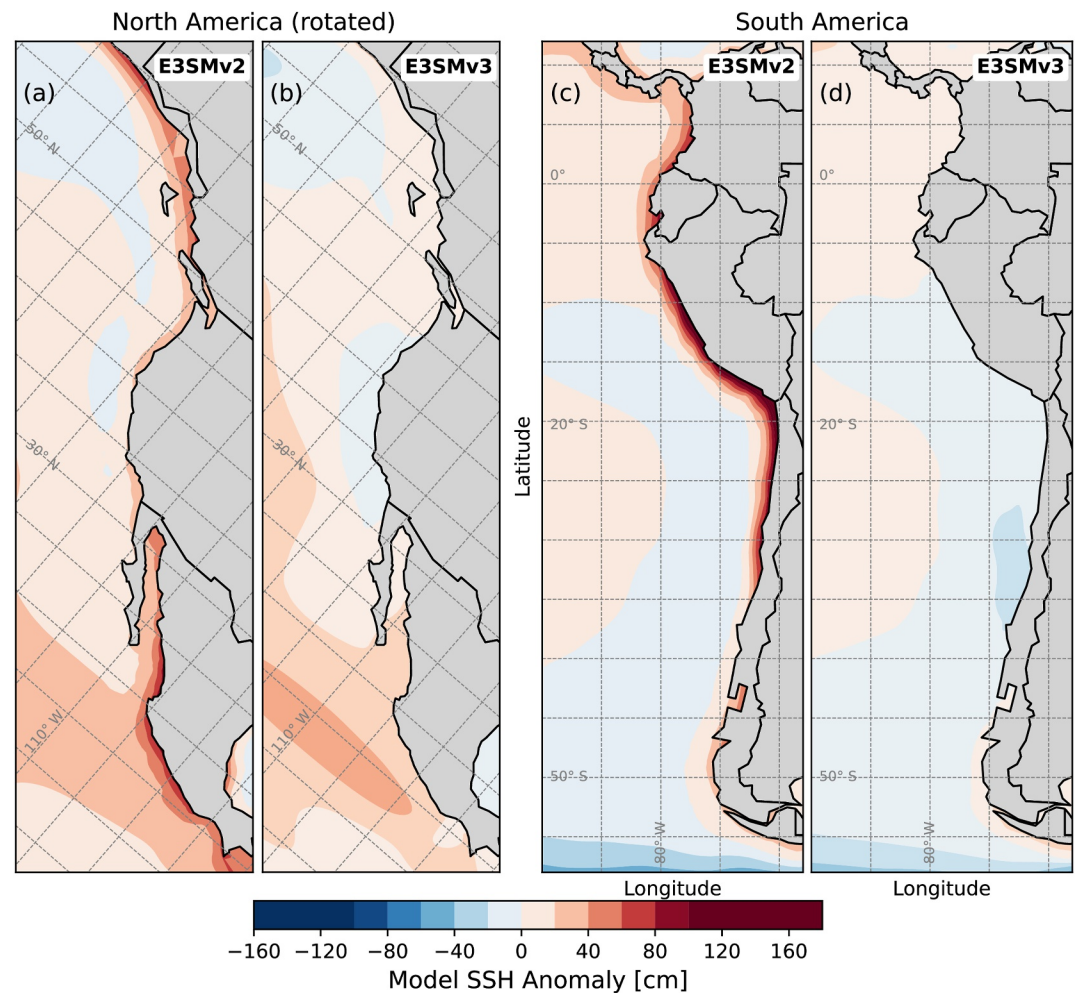


Figure 8. SSH anomaly relative to the global mean along the west coast of North and South America, respectively, in panels (a, c) E3SMv2 *piControl* and (b, d) E3SMv3 *piControl* simulations (years 1–500). In E3SMv3, the correction to the coupling pressure removes the coastal bias that was present in E3SMv2. Note that in panels (a, b) the layout is rotated to display the full extent of the North America coast. Gray dashed lines indicate 5° latitude and longitude lines.

As in previous versions of E3SM, the simulated AMOC remains quite weak relative to observations. Figure 5b shows the maximum AMOC at 26.5°N, coincident with the RAPID/MOCHA mooring array. The 2004–2023 mean AMOC at RAPID is approximately 17 Sv (McCarthy et al., 2025). The two-dimensional stream function shows that the E3SM simulated AMOC is too weak across the entire Atlantic domain (not shown).

The Nino3.4 region index time series is shown in Figure 6a. As in previous versions of E3SM, Nino3.4 variability is strong, with several strong El Niño and La Niña events. To assess the variability relative to previous versions, we subset the control run into five 100-year segments and compute the power spectrum for each segment (Figure 6b). ERSSTv5 data (Huang et al., 2017) is shown in the thick black line and the 90% confidence interval is the dashed black line. Similar to E3SMv1 and v2, there is a pronounced peak of variability near 3 years, which is still slightly less than ERSSTv5, but closer than previous versions. In E3SMv3, longer-term variability in Nino3.4 has again reduced (compare to Golaz et al., 2022, their Figure 16) with only a slight significant peak of variability at 9 years in one of the 100-year segments.

Regional SSTs are also very stable over the entire *piControl* simulation. Figure 7 depicts SST biases relative to a pre-industrial estimate (1870–1900) from HadISST averaged over the first and last 50 years of the *piControl*. Differences between bottom panels are minute, a good indication of a stable climate as desirable for a pre-industrial control simulation.

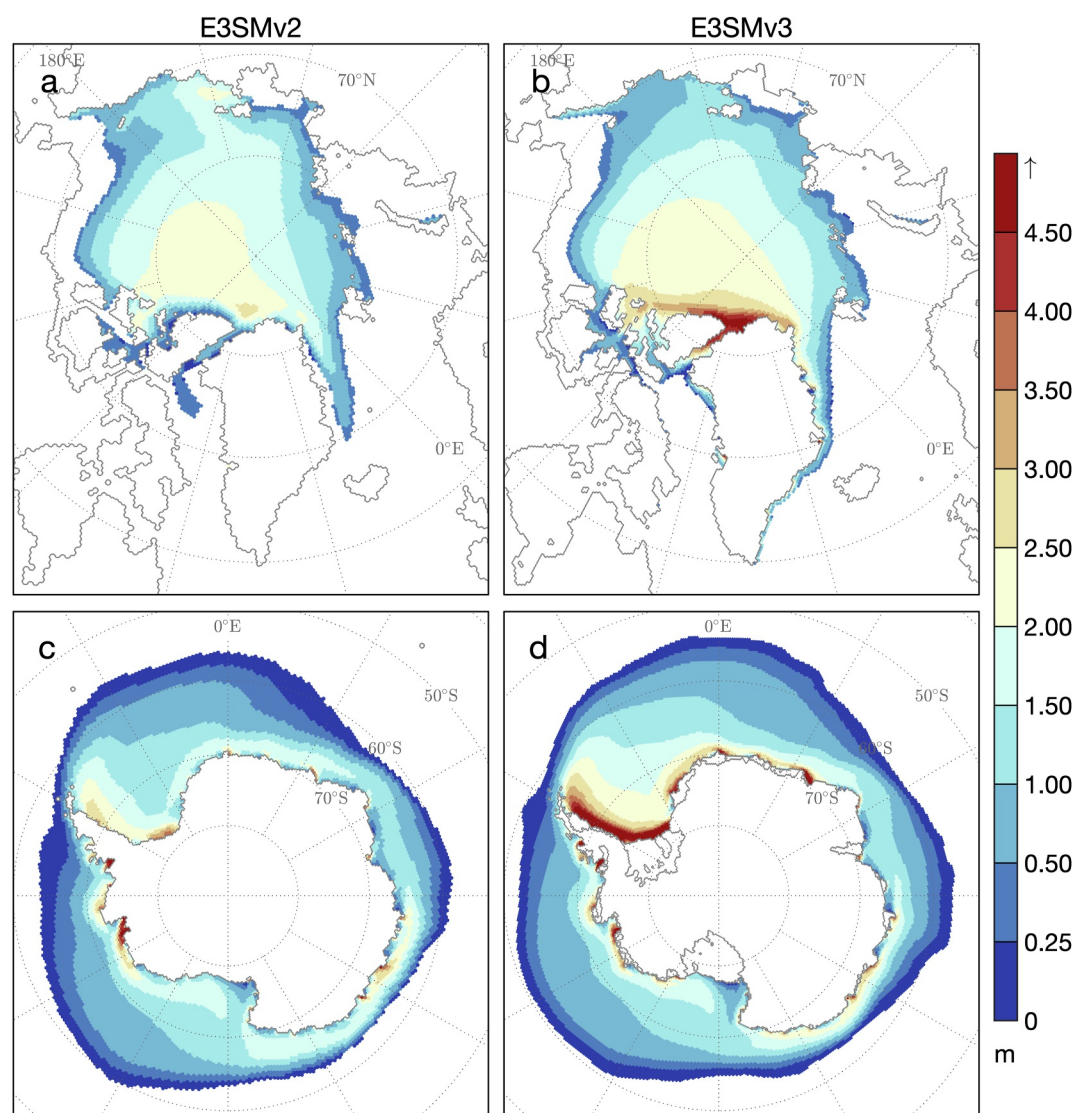


Figure 9. Grid-cell mean 500-year *piControl* September sea ice thickness out to 15% concentration in E3SMv2 (a, c) and E3SMv3 (b, d) illustrating the impact of improved atmosphere-ocean pressure coupling in E3SMv3. Antarctic ice shelves in E3SMv3 (Figure 2) are part of the landmask in panel (d).

Compared to HadISST, E3SMv3 exhibits several well-known regional biases common to coupled climate models (Figure 7). First, the model shows a cold bias in the North Atlantic Subpolar Gyre—a persistent challenge for CMIP5 and CMIP6—characterized by an intermodel spread of 4 psu in sea surface salinity (SSS) and 7°C in SST (Reintges et al., 2024). Consistent with its predecessors E3SMv1.0 and E3SMv1.1, the standard-resolution E3SMv3 falls into the cold–fresh category in this region. Second, E3SMv3 exhibits warm biases in the Southern Ocean, a feature shared by 80% of CMIP6 models (Luo et al., 2023), potentially linked to a weak Atlantic Meridional Overturning Circulation (C. Wang et al., 2014), cloud-radiative feedbacks (Hyder et al., 2018), and/or the influence of a remote warm bias in the North Atlantic deep ocean (Luo et al., 2023). Third, the model displays warm biases in coastal upwelling regions, particularly near Namibia and western North America. These coastal biases are attributed to underestimated shortwave reflection by low-level clouds and unresolved small-scale upwelling processes; for example, convective boundary-layer mixing controls surface evaporative cooling, which can contribute as much as radiative effects (Hourdin et al., 2020). Increasing both oceanic and atmospheric resolution has been shown to substantially reduce near-shore SST biases (Balaguru et al., 2021).

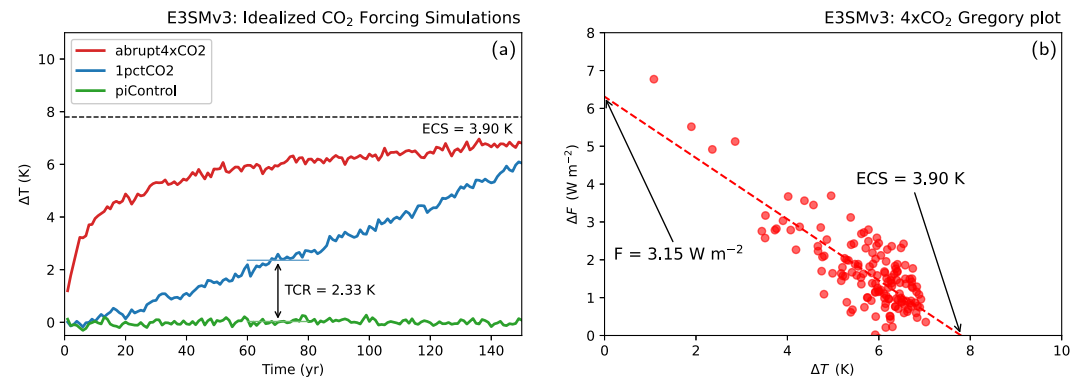


Figure 10. (a) Temporal evolution of annual global mean surface air temperature anomalies for the idealized CO₂ forcing simulations *abrupt-4xCO₂* (red), *1pctCO₂* (blue), and the control simulation (*piControl*; green) for E3SMv3. The transient climate response (TCR) is computed as a 20-year average around the time of CO₂ doubling (year 70). (b) Gregory regression to estimate effective climate sensitivity (ECS) and effective 2xCO₂ radiative forcing (F) using the full 150 years. Anomalies are computed relative to the average of years 101–250 from the underlying control simulation.

There was a major improvement in the 500-year SSH anomaly in E3SMv3 along coastal boundaries as a direct consequence of the switch to mean sea level pressure acting on the ocean surface instead of mid-level surface EAM pressure described in Section 2.6.3. Figure 8 highlights the change in the pressure-adjusted SSH anomaly relative to the simulated global mean in E3SMv2 versus E3SMv3, most visible along the West Coast of North and South America. In previous versions of E3SM, coastal SSH is elevated, up to 40–60 cm along the US, Canadian and Mexican coasts and over 100 cm along the coasts of Peru and Chile. This occurred due to topographic smoothing in EAM reducing surface layer pressure relative to mean sea level pressure near coastlines, thereby elevating coastal SSH. The resulting SSH gradient forced sea ice from coastlines including Greenland, the Canadian Archipelago and Antarctic Peninsula, illustrated in Figure 9, where E3SMv3 better represents coastally constrained sea ice dynamics. The correction of this anomaly significantly impacts global sea ice mass as we will later demonstrate.

4.2. Climate Sensitivity and Transient Climate Response

We estimate the Transient Climate Response (TCR) and ECS of E3SMv3 using a pair of 150-year idealized CO₂ forcing simulations (*1pctCO₂* and *abrupt-4xCO₂* in Table 2). TCR is estimated as the change in surface air temperature for a 20-year period around the time of CO₂ doubling in the *1pctCO₂* simulation. The ECS is defined as the equilibrium surface air temperature change resulting from a doubling in CO₂ concentrations. We approximate it by computing the “effective climate sensitivity” by linearly regressing net TOA radiation versus surface air temperature in the *abrupt-4xCO₂* simulation (Gregory et al., 2004). For E3SMv3, TCR is 2.33 K and ECS 3.90 K (Figure 10). Those values are close to E3SMv2 (TCR = 2.41 K; ECS = 4.00 K) but substantially smaller than E3SMv1 (TCR = 2.93 K; ECS = 5.30 K). E3SMv3 is within one standard deviation of a broad set of CMIP6 models (Meehl et al., 2020), TCR = 2.0 ± 0.4 K and ECS = 3.7 ± 1.1 K, but on the high end of the likely range of 2.6–3.9 K assessed by Sherwood et al. (2020) following multiple lines of evidence.

4.3. Temporal Evolution of Historical Simulations

4.3.1. Trends in Surface Air Temperature

Previous versions of E3SM suffered from the so-called “pothole cooling” problem. While this problem manifests itself in a number of CMIP6 models, the problem is particularly acute in E3SMv1 and E3SMv2. It refers to an anomalous cold bias in surface air temperature simulations during the mid to late 20th century, particularly from 1960 to 1990. This feature is most prominent in the northern hemisphere mid-latitudes and has raised concerns about the accuracy of historical climate reconstructions and the representation of key physical processes in Earth system models.

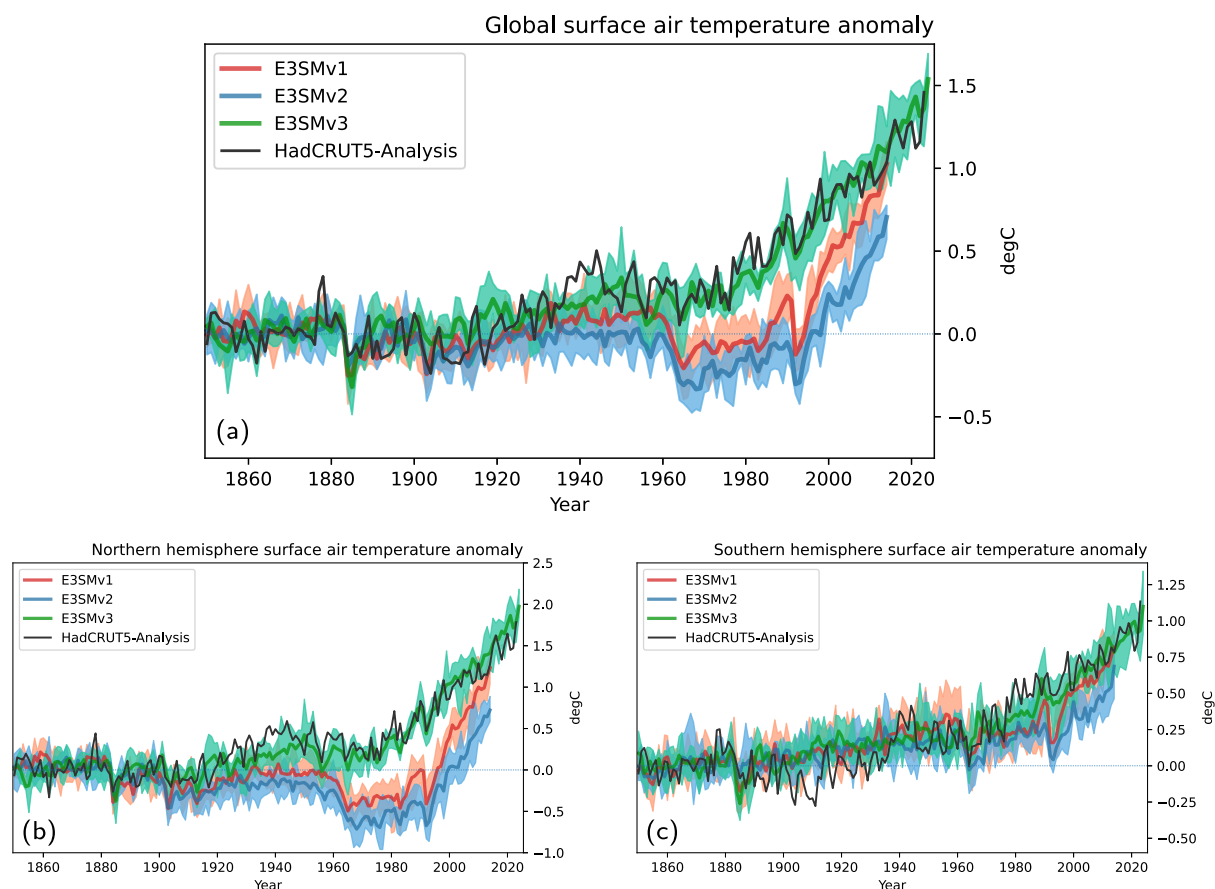


Figure 11. Annual surface air temperature anomalies (with respect to 1850–1899): (a) global, (b) Northern Hemisphere, and (c) Southern Hemisphere. Anomalies are for E3SMv1 (red), E3SMv2 (blue), and E3SMv3 (green) and compared to HadCRUT5-Analysis observational data (black). Lines represent ensemble mean and shading ensemble spread across five historical members. Anomalies are computed separately for each ensemble member and observations by removing the corresponding average over 1850–1899.

Early investigations attributed this cooling primarily to an overestimation of anthropogenic aerosol forcing, especially from sulfate aerosols derived from sulfur dioxide (SO_2) emissions during the post-war industrial boom. These aerosols increase atmospheric albedo via direct scattering and cloud-brightening effects, thus cooling the Earth's surface. J. Zhang et al. (2021) found that structural differences in aerosol–cloud interactions—particularly regarding cloud amount and radiative effects—drive much of the inter-model spread in cooling during this period. However, more recent work by Flynn et al. (2023) challenge the notion that strong aerosol forcing alone can

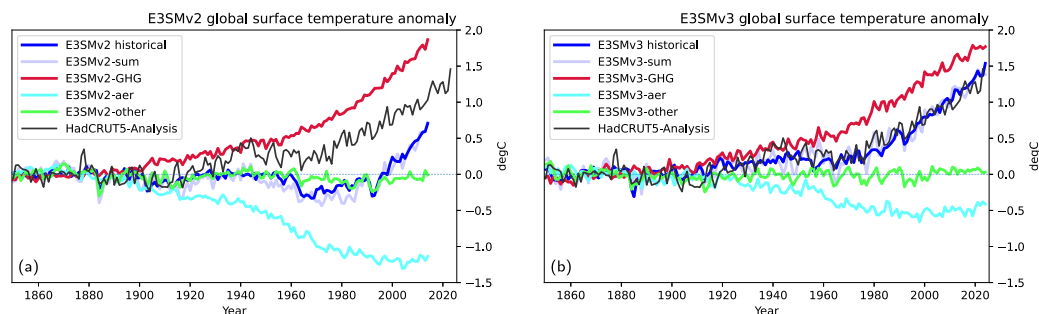


Figure 12. Global surface air temperature anomalies (with respect to 1850–1899) for (a) E3SMv2 and (b) E3SMv3 compared against observations from HadCRUT5-Analysis (black). Comparison of E3SM simulations with all forcings (dark blue), anthropogenic GHGs only (red), aerosol and precursors only (turquoise) and everything else (green). Light purple lines represent the sum of the decomposition.

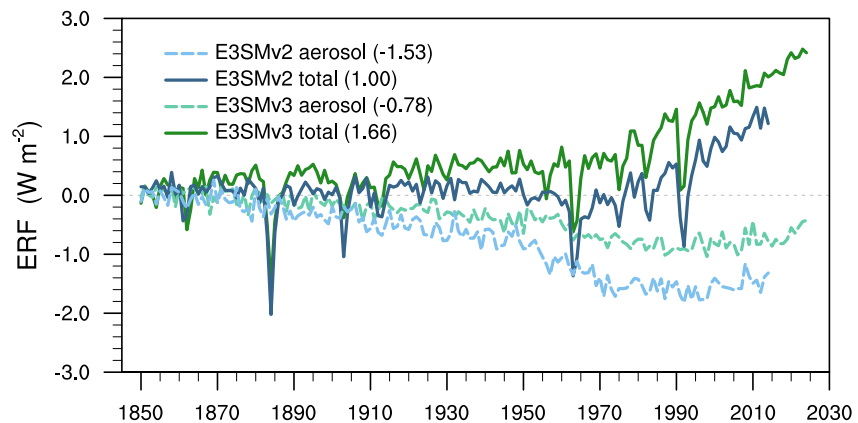


Figure 13. Temporal evolution of global annual mean total (solid) effective radiative forcing (ERF) and aerosol-related (dashed) ERF for E3SMv2 and E3SMv3, respectively. Values enclosed in the parentheses of figure legend denote 20-year average from 1995 to 2014 for each case.

explain the extent of the cold bias. Their analysis of CMIP6 models shows that while models with more accurate mid-century temperatures tend to simulate weaker aerosol cooling, there is no consistent correlation across the ensemble between aerosol forcing strength and the magnitude of the cold bias. This indicates that other physical processes and model characteristics may also contribute substantially to the problem.

However, previous analysis of the relative importance of GHGs warming versus aerosol-related cooling during the historical period revealed that correcting the pothole problem in E3SMv2 would require a very substantial reduction in the magnitude of aerosol effective radiative forcing (ERF_{aer} ; Golaz et al., 2022). As such, reducing the magnitude of ERF_{aer} became a development priority for the atmosphere component of E3SMv3. Table 5 in Xie et al. (2025) lists 10 aerosol and cloud microphysics changes in the atmosphere from E3SMv2 to E3SMv3 and evaluates their impact on the ERF_{aer} . The three greatest contributors to reductions in (the magnitude of) ERF_{aer} were parametric changes that primarily impacted [the effective radiative forcing from aerosol-cloud interaction] ERF_{aci} :

1. Increase in minimum cloud droplet number concentration (Nd) limiter: The prescribed lower bound for Nd was raised from 10 cm^{-3} in EAMv2 to 20 cm^{-3} , weakening ERF_{aer} by $+0.25 \text{ W m}^{-2}$.
2. Doubling of DMS emissions: This increased background cloud condensation nuclei (CCN) concentrations, weakening ERF_{aer} by $+0.12 \text{ W m}^{-2}$.
3. Recalibration of Nd exponent: The autoconversion Nd exponent was changed from -1.4 to -1.1 , weakening ERF_{aer} by $+0.1 \text{ W m}^{-2}$.

The net result is shown in Figure 11. The temporal evolution of surface air temperature anomalies in the E3SMv3 historical ensemble (green) now matches the observed record from HadCRUT5-Analysis (black; Morice et al., 2021) very closely for the global average (Figure 11a) as well as the northern hemisphere (Figure 11b) where the mismatch between E3SMv1 (red), E3SMv2 (blue) and observations was particularly pronounced due to the higher concentrations of anthropogenic aerosols in that hemisphere. The southern hemisphere (Figure 11c) also improves slightly in E3SMv3.

To understand the improvement in the evolution of the global surface air temperature between E3SMv2 and E3SMv3, we repeat an ensemble of coupled simulations spanning 1850–2024 but selectively activating certain time-varying forcing agents as described in Section 3. Results are illustrated in Figure 12. The dominant forcing contributions are GHG (red) and aerosol-related (turquoise). The remaining contributions (green) show inter-annual variations (mostly from volcanic eruptions and the solar cycle) without a long-term trend. A summation over the decomposition (light purple) reproduces the original historical ensemble (dark blue) very well, indicating that the decomposition is complete and linear.

Turning to differences between model versions, there is a slight reduction in the warming from GHGs in E3SMv3 compared to E3SMv2, particularly during 2000–2014. Possible causes include a smaller TCR (2.33 K vs. 2.41 K) in E3SMv3 compared to E3SMv2, and reduced concentration of some GHGs (including tropospheric ozone, CH_4 ,

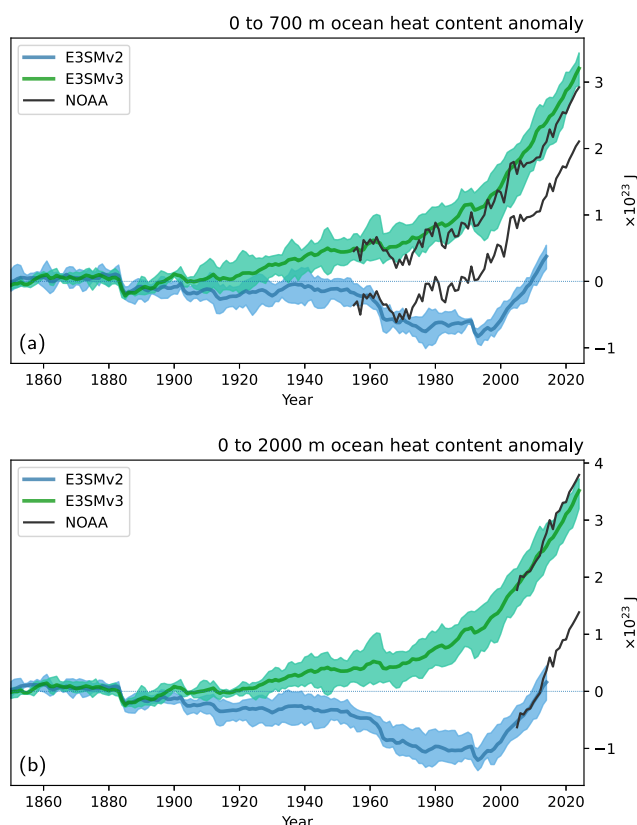


Figure 14. Temporal evolution of annual OHC anomalies for two depth ranges, (a) 0–700 m and (b) 0–2,000 m, compared across E3SMv2 (blue), E3SMv3 (green), and NOAA observational data (black). Model anomalies are computed with respect to 1850–1899. Observations are displayed twice, normalized over the first 10 years to match each model anomalies for consistent comparison. Shaded regions represent the ranges for the E3SMv2 and E3SMv3 historical ensemble.

4.3.2. Trends in Ocean and Sea Ice

It is also noteworthy that the reduction in simulated anthropogenic aerosol cooling has a profound impact in the evolution of OHC as illustrated in Figure 14 for two depth ranges (0–700 m and 0–2,000 m). At both depth ranges, net OHC in E3SMv2 is flat or trending negative between 1850 and 2010 with a pronounced cooling between 1960 and 1990 coinciding with the pothole cooling problem in the atmosphere. In contrast, the trend in OHC in E3SMv3 becomes positive starting around 1920 and rapidly accelerates in the late 20th and early 21st centuries.

Observational estimates of OHC are available from NOAA (https://www.ncei.noaa.gov/access/global-ocean-heat-content/basin_heat_data.html) based on the methodology described in Levitus et al. (2012). The yearly records start in 1955 for 0–700 m and 2005 for 0–2,000 m. To compare observed and simulated trends, the observed record is plotted in Figure 14 as anomalies normalized with respect to the first 10 years for each model. For 0–700 m, the trend simulated by E3SMv2 is extremely unrealistic. E3SMv3 corrects the problem with the observation falling within the five-member ensemble spread from 1955 to 2024. For 0–2,000 m, the record is much shorter, but E3SMv3 also matches the observed rate of warming. We note that starting in 2000, E3SMv3 appears to overestimate the warming in 0–700 m and underestimate it in 0–2,000 m. This is likely driven by the weak AMOC which indicates a lack of vertical transport of heat to depth. The lack of vertical heat transport is also consistent with the overly strong stratification seen in vertical sections in E3SMv3 (not shown) and with the higher TCR in E3SMv3 compared to other models (Hu et al., 2020 and Section 4.2).

Figure 15 summarizes global E3SMv3 1850–2024 evolution of sea ice extent and volume as compared to *piControl*, E3SMv2, OHC, and the NSIDC/NOAA sea ice observational data record of Meier et al. (2025). For

and N_2O) due to the interactive tropospheric gas chemistry and updates of stratospheric chemistry in E3SMv3 (Tang et al., 2025).

But the largest difference between model versions emerges in the aerosol-related cooling. The magnitude of the cooling is substantially reduced from -1.25 K in E3SMv2 to around -0.5 K, which is key to correcting the pothole cooling problem in E3SMv3. To further attribute the change, we also estimate the historical effective radiative forcing (ERF) from 1850 to the present day using the RFMIP simulations (Table 2). Total ERF is derived by calculating the difference in net TOA radiative flux imbalance between the *piClim-histall* (transient) and *piClim-control* (pre-industrial) simulations. Similarly, aerosol-related ERF is quantified by contrasting *piClim-histaer* (transient) with *piClim-control*.

Figure 13 shows the temporal evolution of global annual mean ERF, distinguishing total (solid) and aerosol-only (dashed) forcings in E3SMv3 and E3SMv2, as well as the averaged ERF values over 1995–2014 in the legend. Each line represents an ensemble mean of three members. Both model versions exhibit a negative aerosol-only ERF, indicative of an overall cooling effect from anthropogenic aerosols. However, the magnitude of the forcing is substantially reduced in E3SMv3 (-0.78 W m^{-2}) relative to E3SMv2 (-1.53 W m^{-2}) and E3SMv1 (Golaz et al., 2019; K. Zhang et al., 2022). This is consistent with findings in Xie et al. (2025), which reported a much reduced aerosol forcing in E3SMv3 due to atmospheric model physics schemes and parameter tuning changes related to aerosol and cloud representation.

The total ERF, which includes anthropogenic GHGs, aerosols, and other forcing agents, is positive across both model versions, yet significantly larger in E3SMv3 (1.66 W m^{-2}) than in E3SMv2 (1.00 W m^{-2}). This net forcing increase is mainly due to the reduced aerosol cooling (by 0.75 W m^{-2}). Overall, the results show a reduced magnitude of aerosol ERF and thus an enhanced magnitude of total ERF in E3SMv3, highlighting the fundamental importance of aerosol contribution to the total forcing that drove the historical surface temperature trends, as shown in Figures 11 and 12.

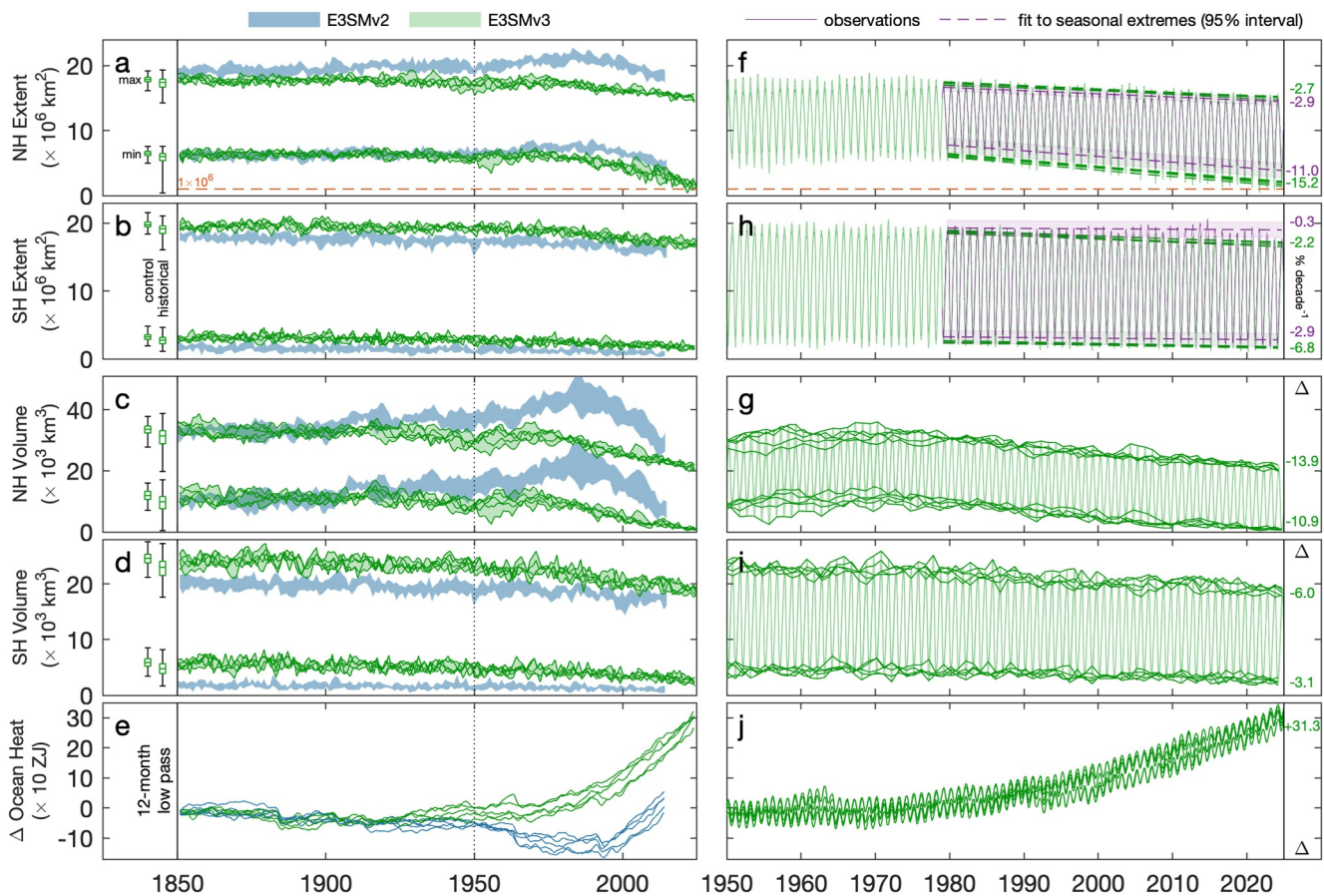


Figure 15. E3SMv2 (blue) and E3SMv3 (green) Northern (a, c; NH) and Southern (b, d; SH) hemisphere sea ice extent and volume extremes from the five ensemble members compared to 12-month filtered OHC (e). Box plots on the left compare daily values of the 500-year *piControl* with the 1850–2024 E3SMv3 five-member historical ensemble. Detailed respective daily ensemble values from 1950 to 2024 appear in the right column (f–j), including observed 1979 to 2024 Meier et al. (2025) sea ice extent (purple), and a linear fit to seasonal extremes, measured and for each E3SMv3 ensemble member (dashed). The 95% standard error spread surrounds the observed trend in transparent purple (f, h). Numbers in the right column indicate seasonal extreme trend of the five-member E3SMv3 ensemble mean and observed extent (f, h; green and purple respectively), 1850–2024 ensemble mean change (Δ) in sea ice volume (g, i) and global heat content (j). The horizontal dashed orange line (a, f) indicates 1 million km^2 extent, one definition of complete Arctic sea ice loss (Jahn et al., 2024), the vertical stippled line (a–e) indicates the 1950 temporal boundary of the right panels (f–j).

this, passive microwave-derived concentration was quality controlled for all full years of the data set, 1979 to 2024. Errant daily Scanning Multichannel Microwave Radiometer (SMMR) concentration fields were removed due to spikes in extent during September 13–17, 1984 (Global), March 29 to April 9, 1986 (Northern Hemisphere), August 5, 1985 and June 19–25, 1986 (Southern Hemisphere). The quality-controlled data set was then used to calculate hemispheric seasonal extreme extent trends shown in purple in the right-most column of Figure 15. Results indicate the E3SMv3 spring Arctic trend lies within observational uncertainty across the ensemble (-2.7 obs vs. -2.9 model $\% \text{ decade}^{-1}$), while the ensemble loses too much sea ice as compared to the daily fall observed minimum (-11.0 obs vs. -15.2 model $\% \text{ decade}^{-1}$). Changes in volume and OHC relative to initial conditions of each ensemble member (Δ) are also shown (Figures 15g–15j).

Regardless of these moderate trend differences, the sea ice improvement in E3SMv3 relative to E3SMv2 is unmistakable (Figure 15). Sea ice extent lies within close range of observations, in contrast to E3SMv2 (blue ensemble spread in Figure 15), and the impact of improved coastal build-up of sea ice discussed earlier can be seen in increased sea ice volume throughout the industrial era Southern Hemisphere volume. There is a notable increase in ensemble spread that coincides with increased volcanic activity in the 1950s, and an increase in volume prior to the steady Arctic decline after 1970.

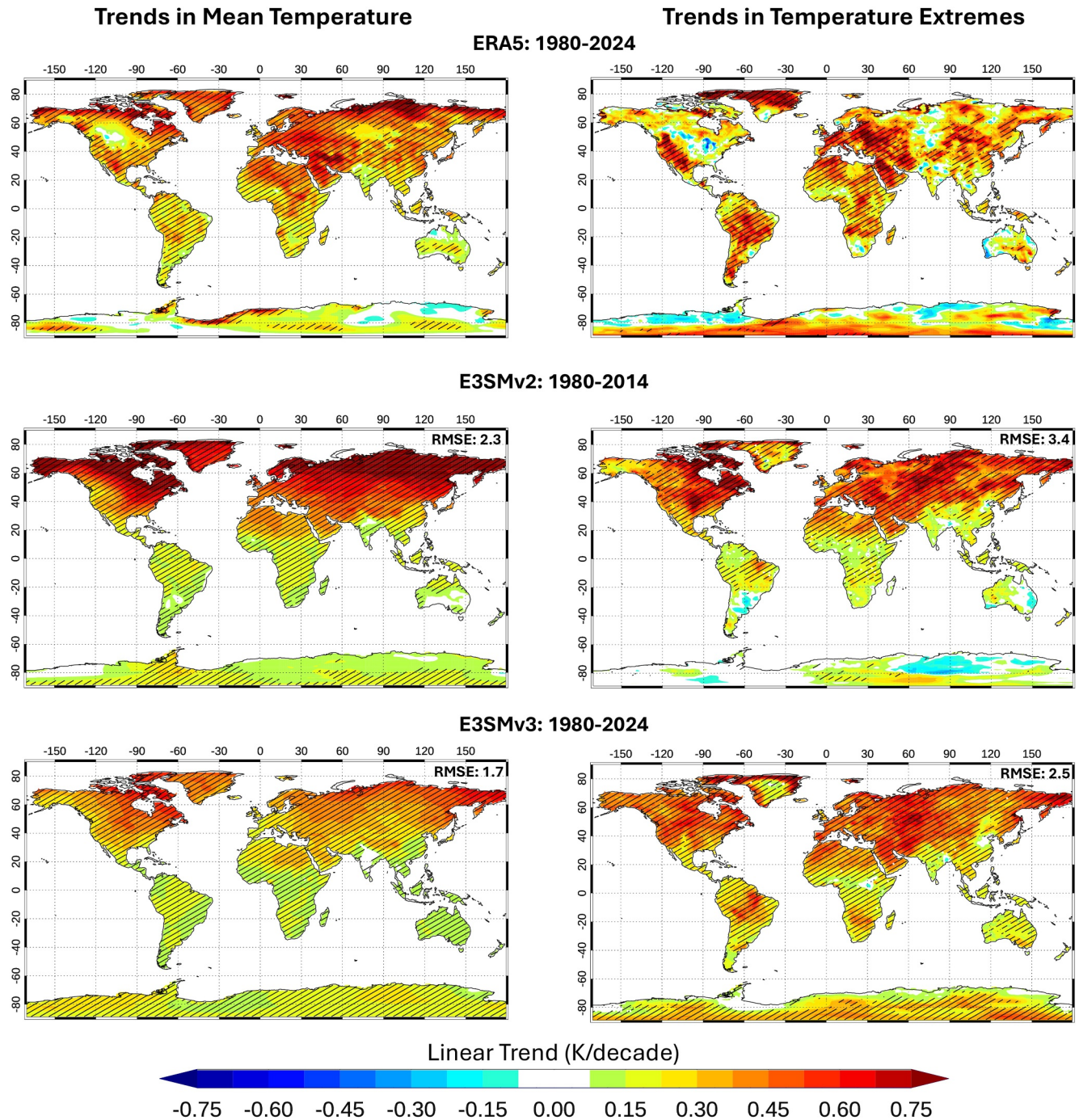


Figure 16. Linear trends in mean and temperature extremes. Left column: linear temporal trend in annual mean reference height (2 m) temperature at each grid point over global land for ERA5 (1980–2024) (top), E3SMv2 five-member ensemble (1980–2014) (middle) and E3SMv3 five-member ensemble (1980–2024) (bottom) computed using linear regression. Right column: Linear temporal trend in extremes as represented by the coefficient (α) of the time co-variate of the location parameter of the generalized extreme value distribution fit at each grid point for the same data sets. Hatching indicates regions where the coefficients are statistically significant at the 95% confidence level. RMSE with respect to ERA5 estimates is also listed for E3SMv2 and E3SMv3. Because return levels are linearly proportional to the location parameter, α can be directly interpreted as the common linear trend across all return levels (e.g., 20-, 50-, 100-year return period extremes).

4.3.3. Recent Trends in Land Near-Surface Air Temperature Extremes

Figure 16 shows the linear temporal trend in annual mean reference height (2 m) temperature for ERA5 (1980–2024), E3SMv2 (1980–2014), and the E3SMv3 (1980–2024) five-member ensemble mean at each grid point.

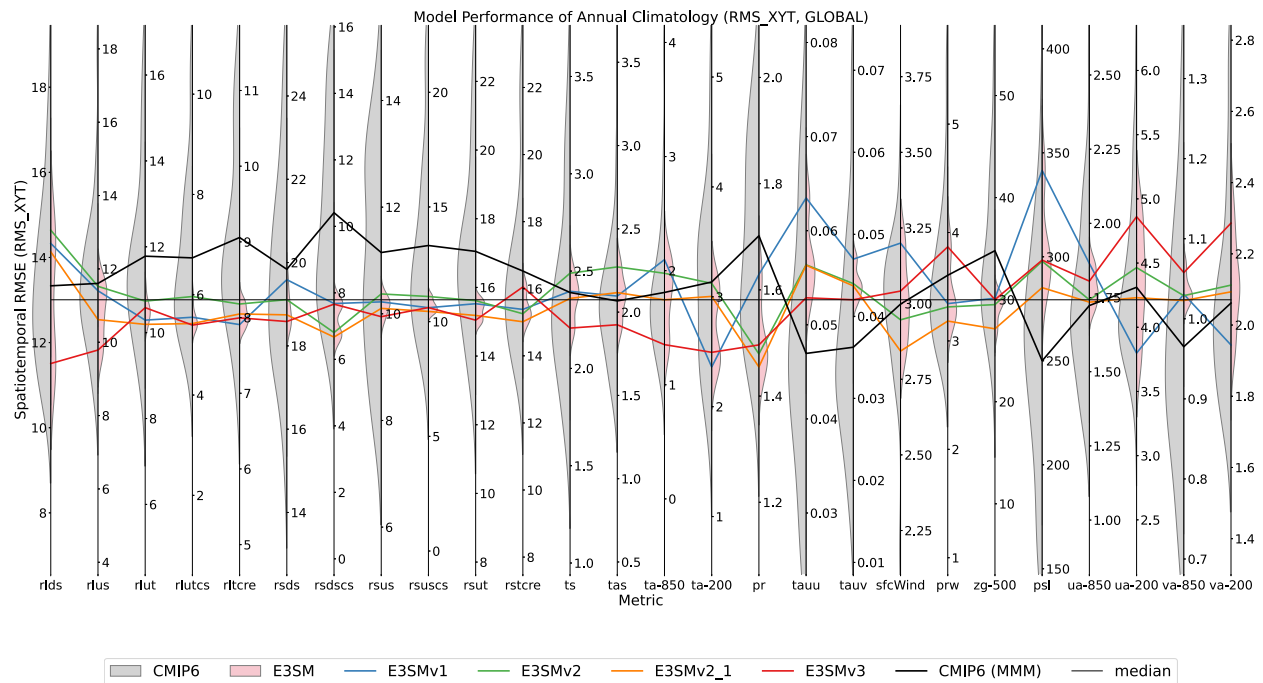


Figure 17. Parallel coordinate plot comparing the performance of E3SMv3 and CMIP6 (first historical members, r1i1p1f1) coupled historical simulations for the annual cycle climatology of multiple model variables (columns), including surface downwelling longwave radiation (rlds; unit: W m^{-2}), surface upwelling longwave radiation (rlus; unit: W m^{-2}), TOA outgoing longwave radiation (rlut; unit: W m^{-2}), TOA outgoing clear-sky longwave radiation (rlutcs; unit: W m^{-2}), cloud radiative effect on longwave radiation (rlutcre; unit: W m^{-2}), surface downwelling shortwave radiation (rsds; unit: W m^{-2}), surface downwelling clear-sky shortwave radiation (rsdscs; unit: W m^{-2}), surface upwelling shortwave radiation (rsus; unit: W m^{-2}), surface upwelling clear-sky shortwave radiation (rsuscs; unit: W m^{-2}), TOA outgoing shortwave radiation (rsut; unit: W m^{-2}), cloud radiative effect on shortwave radiation (rstcre; unit: W m^{-2}), surface temperature (ts; unit: K), near-surface air temperature (tas; unit: K), air temperature at 200 and 850 hPa (ta-200 and ta-850; unit: K), precipitation (pr; unit: mm day^{-1}), zonal and meridional surface wind stress (tauu and tauv; unit: Pa), surface wind speed (sfcWind; unit: m s^{-1}), precipitable water (prw; unit: kg m^{-2}), 500 hPa geopotential height (zg-500; unit: m), sea level pressure (psl; unit: Pa), and zonal and meridional wind components at 850 and 200 hPa (ua-850, ua-200, va-850, va-200; unit: m s^{-1}). CMIP6 native model resolutions range from 0.25° to 2.8° . The E3SMv3 falls in the upper tier. All model and observational data are regridded to $2.5^\circ \times 2.5^\circ$ grids for the comparison. The spatiotemporal RMSE is calculated using the RMSE for each month of the annual cycle and averaging over the months, following the definition from Gleckler et al. (2008). The lines are for the CMIP6 multi-model means (MMMs) and five-member ensemble mean of each E3SM version. The shadings to the left and right of each vertical axis represent the spread of CMIP6 models and the E3SM model family, respectively. The mean climatologies of the reference observational and reanalysis data sets are derived from CERES-EBAF Ed4.1 (Loeb et al., 2018) (2001–2014) for radiation at the surface and top of the atmosphere (r^*), GPCP2.3 (Adler et al., 2018) (1985–2014) for precipitation (pr), and ERA5 (Hersbach et al., 2020) (1985–2014) for all other quantities. Regarding the clear-sky flux estimates, we employed radiative transfer–based calculations from the CERES-EBAF Ed4.1 data set for consistency with the model-derived quantities. Further detailed descriptions of the parallel coordinate framework can be found in J. Lee et al. (2024).

Global average temperatures exhibit strong temporal trends both in observations and simulations after 1980 (Figure 11a). Hatching indicates regions where the trends are statistically distinguishable from zero at the 95% confidence level based on a two-tailed Student's t -test with false discovery rate corrections applied (e.g., Ventura et al., 2004). ERA5 exhibits statistically significant trends globally with stronger trends over large parts of North America, Europe, Northern Asia and Northern and central Africa, consistent with prior studies (e.g., Gulev et al., 2021; Scafetta, 2023).

E3SMv2 simulates stronger than observed trends over large parts of the Northern hemisphere mid- and high-latitudes. The trends are consistent with its stronger than observed temporal trends in Northern-hemisphere average temperatures after 1980 (Figure 11b). However, E3SMv2 exhibits weaker than observed trends over the sub-tropical regions, for example, over Northern and Central Africa, Amazon rainforests, the Middle East and the Southwestern US. E3SMv3, with closer to observed global average temperature trends (Figure 11), simulates weaker trends as compared to E3SMv2, reducing the RMSE with respect to ERA5 trends. E3SMv3 also generally simulates weaker than observed trends over many regions, for example, over the Southwestern US, Northern Africa, Europe, Northern Asia and central South America. Normalized trends, computed as linear trends divided by the inter-annual standard deviation of the detrended time-series at each grid point, highlight tropical amplification as opposed to Arctic amplification both in observations and ESMs (Zeng et al., 2021). E3SMv3 ensemble

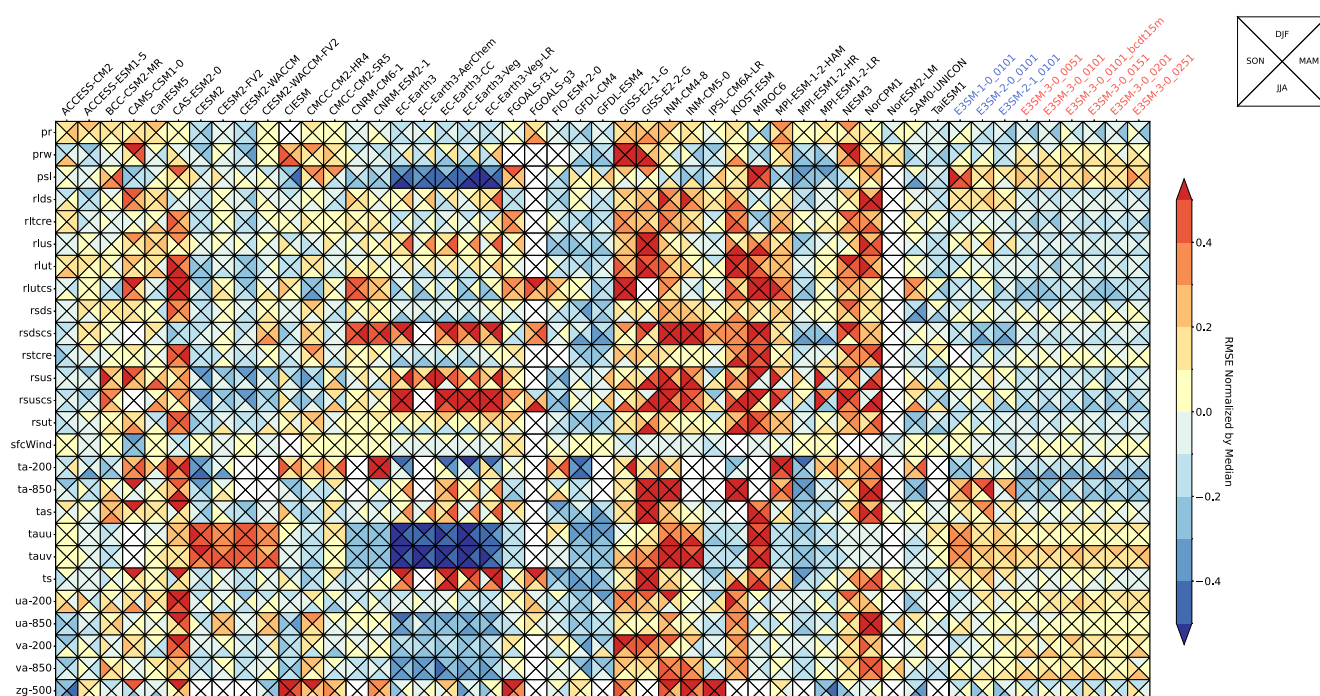


Figure 18. Portrait Plot that compares performance of E3SM against previous versions and CMIP6 (first historical members r11p1f1) coupled historical simulations (columns) for seasonal climatology of multiple model variables (rows) as described in Figure 17. All metrics were derived with the 30-year climatology using the monthly model output from coupled historical simulations during 1985–2014. The Portrait Plot was generated by the PCMDI Metrics Package (J. Lee et al., 2024) that is implemented in the E3SM zppy workflow (C. Zhang et al., 2022). The RMSE was calculated for each season (shown as triangles in each box) over the globe, including both land and ocean, using model and reference data that were interpolated to a common $2.5^{\circ} \times 2.5^{\circ}$ grid. The mean climatology of the reference observations and analysis data sets used to compute the metrics are the same as those in Figure 17. The RMSE of each variable is normalized by the median RMSE of all models. A result of 0.2 (–0.2) indicates an error of 20% greater (smaller) than the median RMSE across all models. White boxes indicate missing values due to variables not being available in the standard CMIP6 post-processed data pool. Further detailed descriptions of the Portrait Plot can be found at J. Lee et al. (2024).

exhibits tropical amplification in normalized trends (not shown). However, it is weaker than ERA5, given the simulation of weaker raw trends in the tropical region in the model. E3SMv2 also indicates stronger normalized trends than raw trends over the tropics. However, normalized trends in E3SMv2 remain strong in the NH high latitudes, likely due to the simulation of stronger than observed raw trends in high latitudes.

Figure 16 also shows estimated trends in extremes of temperature. These are computed by modeling annual maximum temperatures (z) using a generalized extreme value (GEV) distribution, which is represented as a three parameter distribution, $G(z) = GEV(\mu, \sigma, \xi)$, separately at each grid point, where μ, σ , and ξ refer to the location, scale and shape parameters respectively of the distribution (Coles, 2001). Using a generalized linear modeling approach with a linear dependence on time, t , is included as a co-variate in the location parameter ($\mu = \mu_0 + \alpha \cdot t$) when fitting the GEV model, where α quantifies the time-dependence of μ . Positive values of α indicate a rightward shift of the GEV distribution implying an increased likelihood of stronger extremes. All four parameters are estimated using the maximum likelihood approach (Coles, 2001). A GEV model is also fit for each grid point without time as a co-variate, which is treated as the base model. The statistical significance of α is established using a deviance statistic approach, where the test statistic (D) is based on the difference in the log-likelihood of the time-dependent GEV model and the base GEV model (Coles, 2001). We also apply false discovery rate corrections to account for multi-testing following previous studies (e.g., Whan & Zwiers, 2017). This GEV-based approach is commonly used for quantifying non-stationarity in climate extremes (Brown et al., 2008; Evans et al., 2014; Mahajan et al., 2015, 2018; Whan & Zwiers, 2017). The more familiar metric of return level, z_T , for a fixed return period, T , of extremes can also be computed by inverting the GEV distribution and is linearly proportional to the location parameter (Coles, 2001). Thus, α can be directly interpreted as the common linear trend across all return levels (e.g., 20-, 50-, 100-year return period extremes). For model ensembles, α is computed for each ensemble member separately and the ensemble average of α is shown in Figure 16.

ERA5 exhibits statistically significant positive trends in extremes over the Western US, and large swaths of South America, Greenland, Africa, Europe and Central Asia. Most of these regions also exhibit positive trends in mean temperature. The E3SMv2 ensemble broadly shows positive trends over much of the globe. It exhibits stronger than observed extremes over regions where the trend in mean temperature is also stronger, for example, over Eastern and Northern North America and Central and Northern Asia. Over regions where the trends in mean temperatures are weaker than observed, the trends in extremes are also generally weaker for example, over the Western US, South America, Northern and Central Africa and the Middle East. Central Europe also exhibits weaker trends.

E3SMv3 reduces the RMSE of α as compared to E3SMv2 with respect to ERA5 estimates. It also broadly simulates positive trends in extremes over global lands but, consistent with its trends in mean temperature, its simulation of trends in extremes are weaker than E3SMv2. It simulates weaker than observed trends over the Western US, Central Europe, South America, Africa and the Middle East. However, E3SMv3 broadly exhibits stronger than observed trends (although weaker than E3SMv2) over Northern high latitudes, where ERA5 data exhibits weak trends that are not statistically significant. Over the central US, where ERA5 exhibits negative or weak trends in temperature extremes (often referred to as the “warming hole,” e.g., Eischeid et al., 2023). E3SMv3 simulates statistically significant positive trends. E3SMv2 simulates stronger trends there. Recent observational and modeling studies (e.g., Eischeid et al., 2023) suggest that the trends in temperature extremes over the warming hole are associated with anomalously high summertime rainfall there driven by changes in SST patterns. Natural variability of the atmosphere-ocean coupled system (e.g., Meehl et al., 2015) and land use changes (e.g., L. Chen & Dirmeyer, 2020) have also been found to play a role. We will investigate the role of these processes in generating E3SM model biases there in the future. Also, although weaker than observed, E3SMv3 simulates stronger extremes over Central South America and Southern Africa than E3SMv2, even though the trends in the mean are equally weak in the models.

To summarize, we find that E3SMv2—with its stronger trend in global and northern hemisphere mean temperatures post 1980—simulates stronger extremes in Northern mid- and high-latitudes, but it underestimates trends in mean and extreme temperatures over the sub-tropical regions. E3SMv3 simulates weaker trends in extremes over the Northern high latitudes, but stronger trends in extremes over some parts of the sub-tropics. In the future, we plan to analyze a large ensemble of E3SM simulations as well as multi-model ensembles to more broadly assess trends in extremes over both land and oceans.

4.4. Present-Day Climate

Figures 17 and 18 present an overview of model performance across 27 selected atmospheric variables, comparing the coupled historical simulations from E3SM (green, orange, and red lines in Figure 17 and rightmost columns in Figure 18) against 37 CMIP6 models. The evaluation focuses on the climatological representation, using RMSE metrics computed by the PCMDI Metrics Package (J. Lee et al., 2024). RMSE values are calculated for both annual (Figure 17) and seasonal (Figure 18) climatologies over the 1985–2014 period. Reference observations and reanalysis data sets were collected and provided by the E3SM zppy workflow (C. Zhang et al., 2022), covering the same climatological period, except for the radiative fluxes (denoted by r^* quantities) where observational data were only available for 2001–2014.

As shown in Figure 17, the E3SM family of models generally demonstrate competitive performance, with lower bias compared to many CMIP6 models across most surface and upper-atmosphere variables. Notably, E3SMv3 exhibits high skill in simulating the global climatology of surface radiative fluxes, total precipitation, and air temperature at both surface and upper levels. The RMSE values for these variables are smaller than those of the CMIP6 multi-model means (MMMs), as indicated by red lines versus black lines in Figure 17. In contrast, the mean climatology of dynamical fields—such as sea level pressure (psl), and zonal and meridional winds at 200 hPa and 850 hPa (ua-200, ua-850, va-200, va-850)—simulated by E3SMv3 show higher errors than those of the CMIP6 MMMs. However, it is important to note that many CMIP6 models also exhibit substantial spread in these fields, as illustrated by the wide gray shading, highlighting the common challenge of accurately representing large-scale atmospheric dynamics.

E3SMv3 shows overall skill comparable to that of E3SMv2 in reproducing the global climatology of atmospheric fields at the surface and upper troposphere (Figure 17). Notably, the transition from E3SMv2 to E3SMv3 resulted in a systematic reduction of biases in surface radiative fluxes, particularly during the DJF and MAM seasons

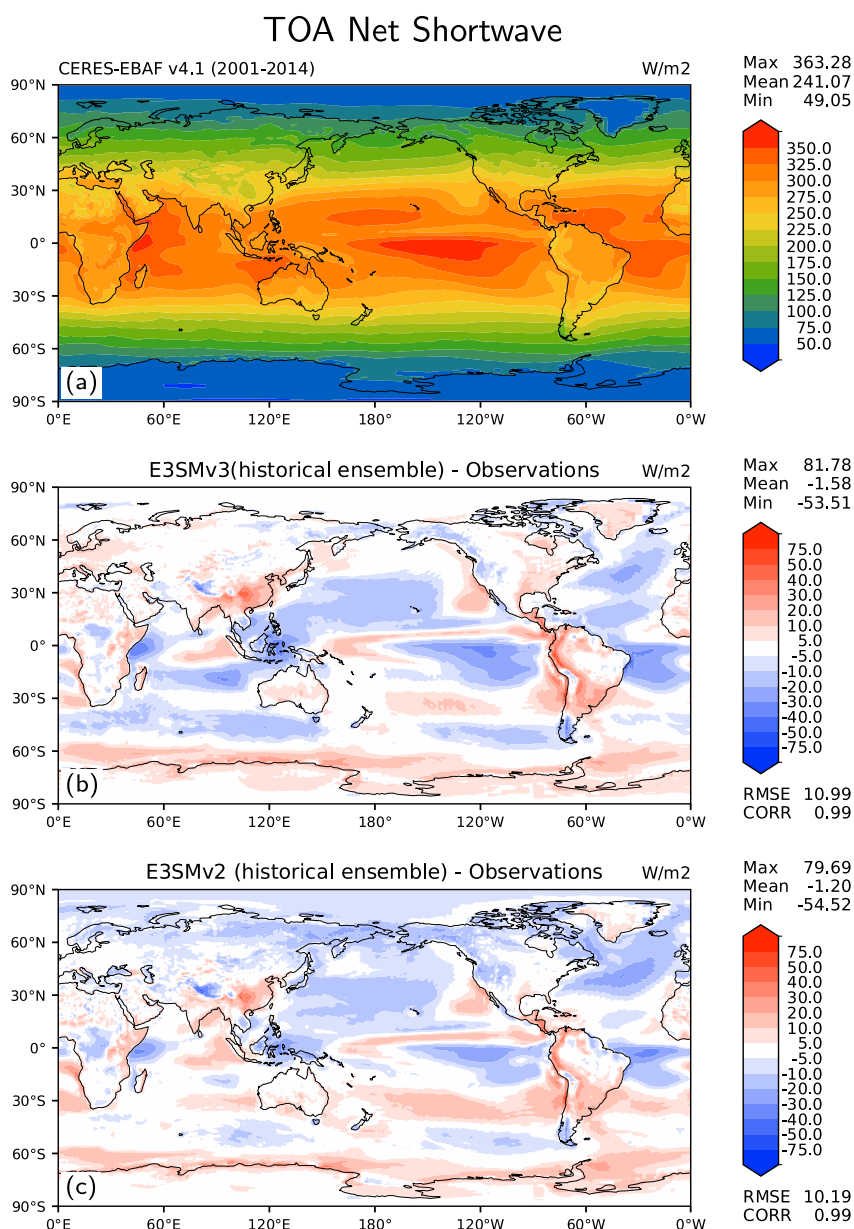


Figure 19. 2001–2014 mean climatology TOA net shortwave fluxes from (a) CERES-EBAF-v4.1, and mean biases in (b) E3SMv3 and (c) E3SMv2.

(Figure 18). These improvements are accompanied by marked enhancements in the simulation of air temperature at the surface and upper levels during the same seasons. However, a slight degradation is observed in E3SMv3 relative to E3SMv2 in the representation of near-surface and upper-level wind fields. Additionally, relatively large biases in mean sea level pressure (psl)—as previously reported in the literature (e.g., Golaz et al., 2022)—remain largely unchanged between the two model versions. These persistent deficiencies highlight areas that warrant further investigation and targeted improvements in future model development efforts. It is also noteworthy that the intermediate version E3SMv2.1 obtained across the board improvements compared to E3SMv2 after AMOC was improved. Figure 18 also demonstrates that reducing the ocean baroclinic time step from 30 to 15 min has very little impact on the model performance (comparing columns on the right denoted E3SM-3-0_0101 and E3SM-3-0_0101_bcdt15m).

We complement this broad overview of the performance of E3SMv3 with a few selected fields, noting that differences between the simulated climatology of E3SMv3 and E3SMv2 are small. Figure 19 shows TOA net

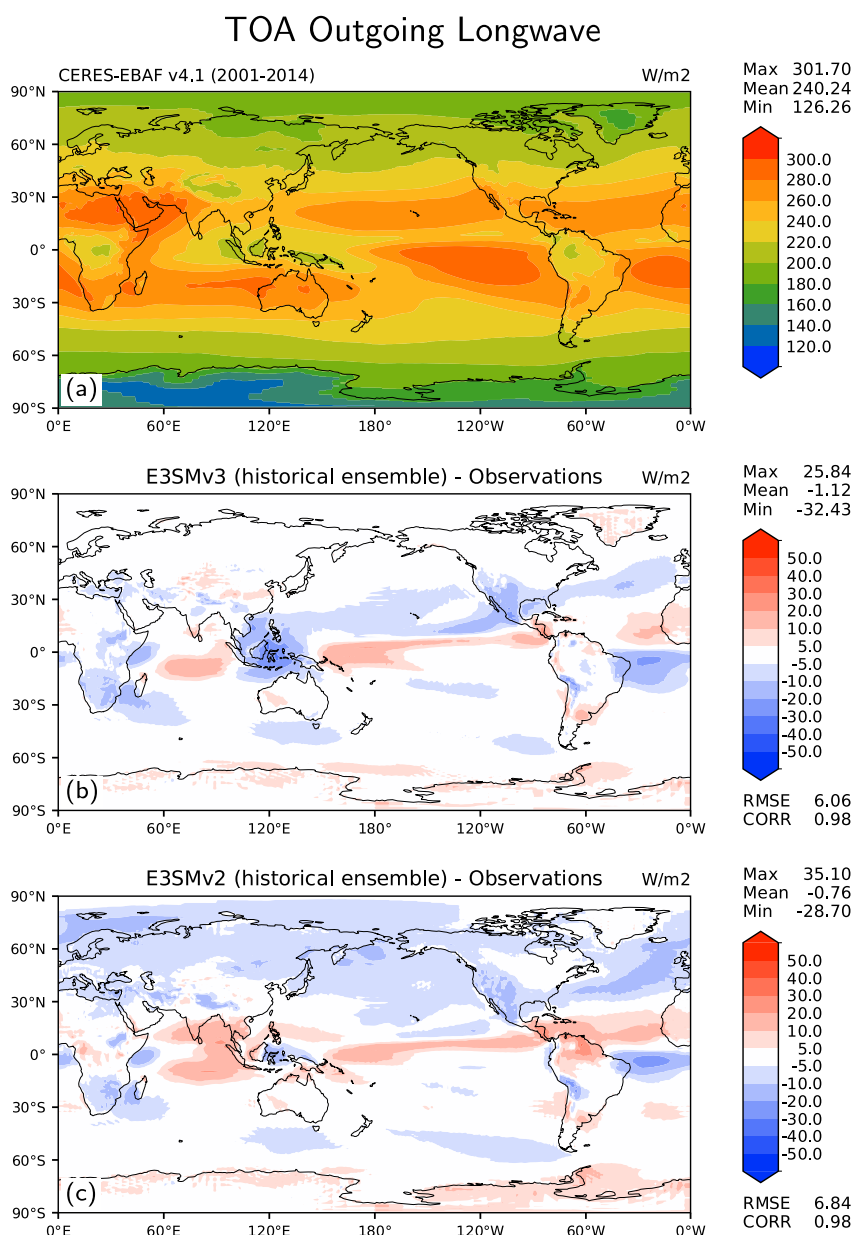


Figure 20. 2001–2014 mean climatology TOA net longwave radiative fluxes from (a) CERES-EBAF-v4.1, and mean biases in panel (b) E3SMv3 and (c) E3SMv2.

shortwave comparing E3SMv2,v3 to CERES-EBAF observations (Loeb et al., 2018, 2024). There is a small degradation as measured by the mean bias and RMSE. Regionally, there is a slight increase in excessive shortwave reflection over the sub-tropical Pacific, but a reduction in northern high latitudes. Outgoing longwave (Figure 20) is slightly improved in E3SMv3 with a reduction in biases over the northern mid- and high latitudes, leading to a slight decrease in RMSE.

The simulation of precipitation is influenced by a number of major changes to the atmospheric physical processes (Section 2.2). The use of P3 cloud microphysical scheme in place of MG2 alters the treatment of large-scale stratiform precipitation process. Several major enhancements to the ZM deep convective scheme not only modulate the triggering of the convection but also change the entire microphysical processes that govern the convective precipitation. Xie et al. (2025) has shown the enhancements to the ZM scheme led to substantial improvements to the simulation of diurnal precipitation with prescribed SST. These improvements remain intact

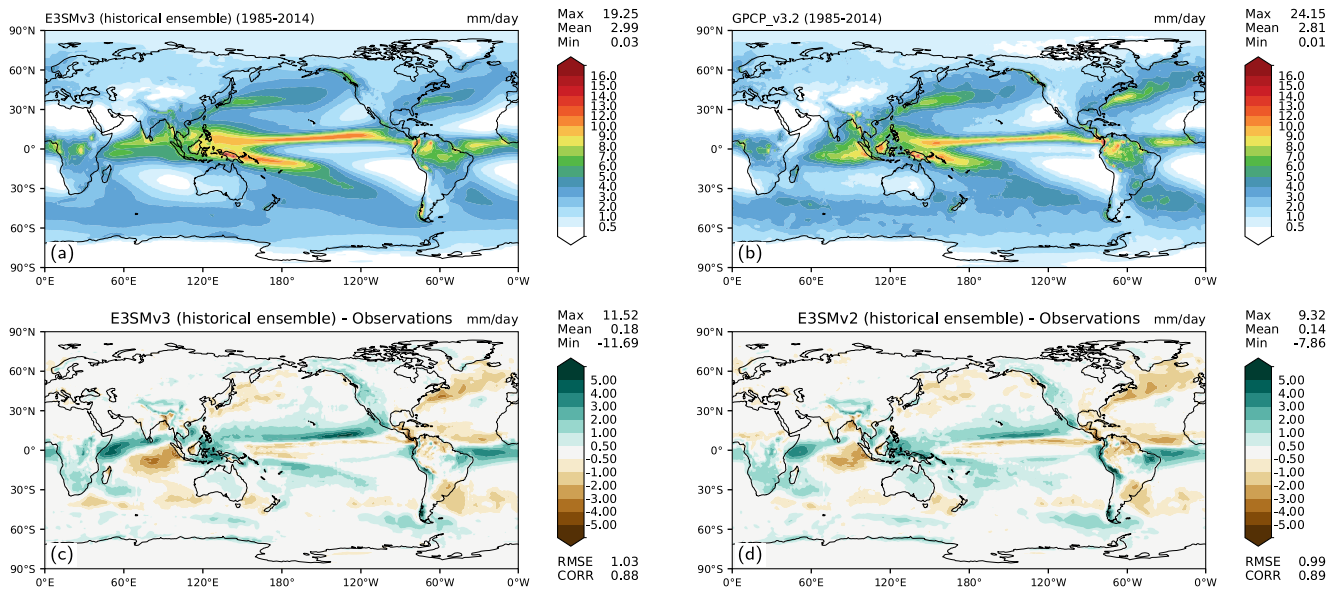


Figure 21. 1985–2014 annual mean precipitation climatology from (a) E3SMv3 and (b) GPCP-v3.2, and biases in panel (c) E3SMv3 and (d) E3SMv2.

in the fully coupled E3SMv3 simulation (not shown). Figure 21 shows the ensemble annual mean precipitation climatology in E3SMv3 mostly tracks the E3SMv2 simulation, with similar spatial bias patterns over the tropical oceans, along the mountainous west coast of North and South America and the North Atlantic storm track. Nevertheless, there are still noticeable differences between E3SMv3 and E3SMv2, which are much more significant compared to the AMIP simulations (see Xie et al., 2025) due to coupled air-sea interactions. The excessive precipitation over the North Pacific Intertropical Convergence Zone (ITCZ) becomes more pronounced, so do the wet biases over the tropical Western Pacific and Northwestern Indian Ocean. Dry biases over North Eastern Indian Ocean where the observed Indian Ocean ITCZ is situated become more severe. There are also several noteworthy improvements from E3SMv2 to E3SMv3. While E3SMv2 had a very mild double ITCZ bias in Central to Eastern Pacific compared to CMIP6 multiple-model ensemble (Tian & Dong, 2020), the biases are further reduced in E3SMv3. Annual mean Atlantic ITCZ was shifted southward in both E3SMv2 and E3SMv3, but it is less severe in the latter. Although E3SMv3 continues to have less precipitation along the Atlantic storm track, the improvement over E3SMv2 is also clear, which to large extent may be attributable to the improved simulation of North Atlantic surface waters (see below). Additional improvements are seen over broader tropical land masses and continental United States. The dipole structure of biases over the Amazon in E3SMv2 is measurably reduced. Similar level of improvements can be seen on the wet biases over Australia and South Africa and dry biases over the equatorial Africa and central United States. Quantifying the contributing factors to these improvements over the lands will need further examination, as the enhancements to the ZM scheme and the prognostic dynamic vegetation model in ELMv3 could all influence the precipitation processes in these regions where convective conditions prevail.

Zonal mean temperature (Figure 22) is generally improved in E3SMv3 consistent with ta-850 and ta-200 in Figure 18, except for an increase in warm bias over the northern mid- and high-latitudes lower stratosphere. While very similar, the zonal mean wind (Figure 23), shows a slight degradation at all heights, consistent with Figure 18.

We also evaluate the historical simulations on their ability to reproduce the observed mean ocean near-surface climate. The observational estimates have distinct integration periods depending on the variable considered: the climatologies are based on the merged Hadley Center-NOAA/OI data from 1990 to 2011 for SST (Hurrell et al., 2008), the NASA Aquarius satellite from 2011 to 2015 for SSS (Lagerloef et al., 2015), the merged absolute dynamic topography satellite data provided by AVISO (Archiving, Validation and Interpretation of Satellite Oceanographic data; processed by SSALTO/DUACS and distributed by AVISO+ (<https://www.aviso.altimetry.fr>) with support from CNES) from 1992 to 2010 for SSH, and ARGO floats from 2000 to 2017 for the mixed layer depth (MLD; Holte et al., 2017). The model output and observations are regridded to a common $0.5^\circ \times 0.5^\circ$ regular grid, and the spatial correlations and RMSE between simulations and observational

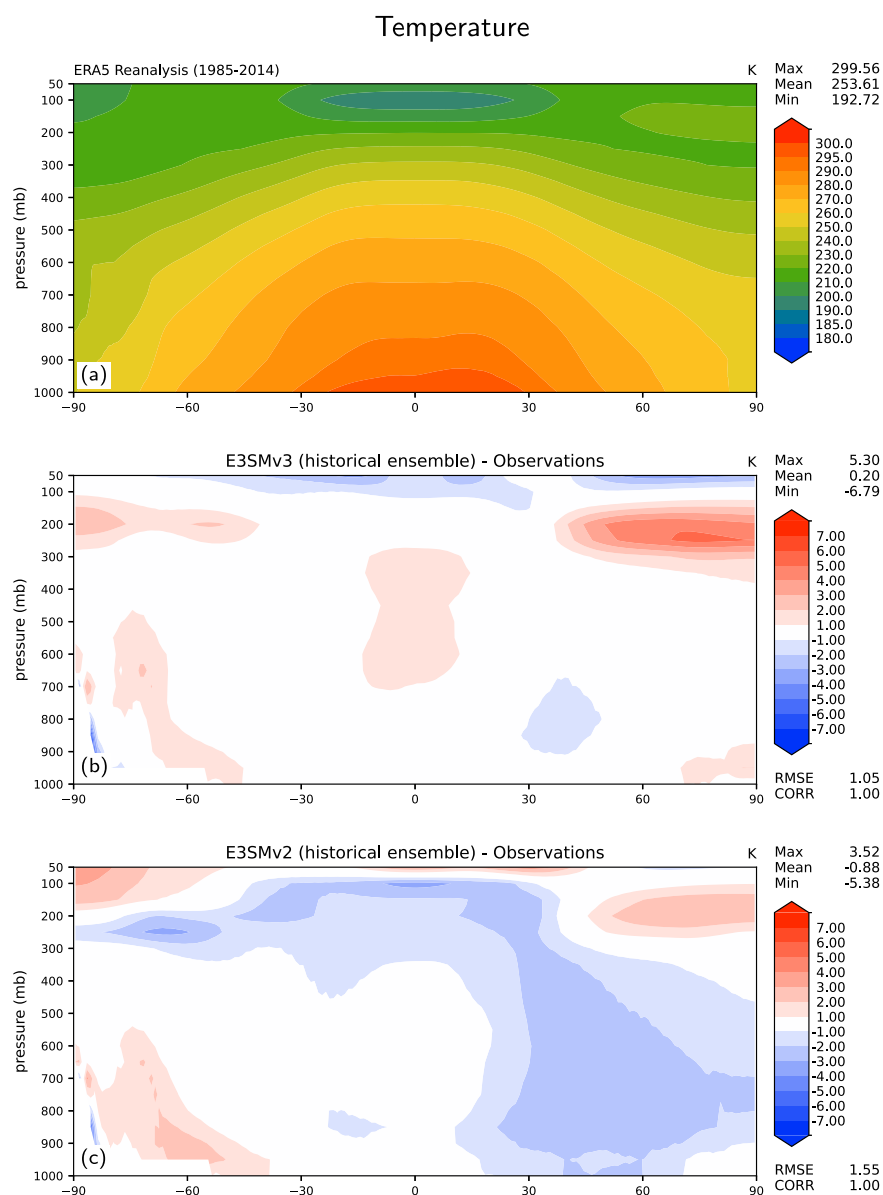


Figure 22. Vertical distribution of zonal mean temperature of 1985–2014 climatology from (a) ERA5, and mean biases in panels (b) E3SMv3 and (c) E3SMv2.

estimates, plotted in Figure 24, are calculated over the model years that overlap the observational period. Spatial correlations between v3 (green) and the observations are systematically higher than v2 (blue) and v2.1 (orange), showing an increased fidelity of the new model version to the observed climate. Likewise, the decrease in RMSE shows an improved representation of the observed climatological values of MLD, SST, and SSS in v3. One exception is the SSH field. While the spatial patterns of SSH have improved from v2 to v3, indicating improved barotropic ocean currents, a net offset in mean global SSH leads to marginally higher RMSE in v3 compared to v2. The additional v3 simulation with a 15 min baroclinic time step (Section 3, marked by a green + in Figure 24) does not exhibit significant differences in the correlation and RMSE metrics, but sits within the multi-realization spread.

As noted in Section 2.3, the ocean changes from v2.1 to v3 include a new mesh and time-stepping method but no changes to the ocean dynamics and thus are unlikely to explain the change in ocean surface climate. Instead, these

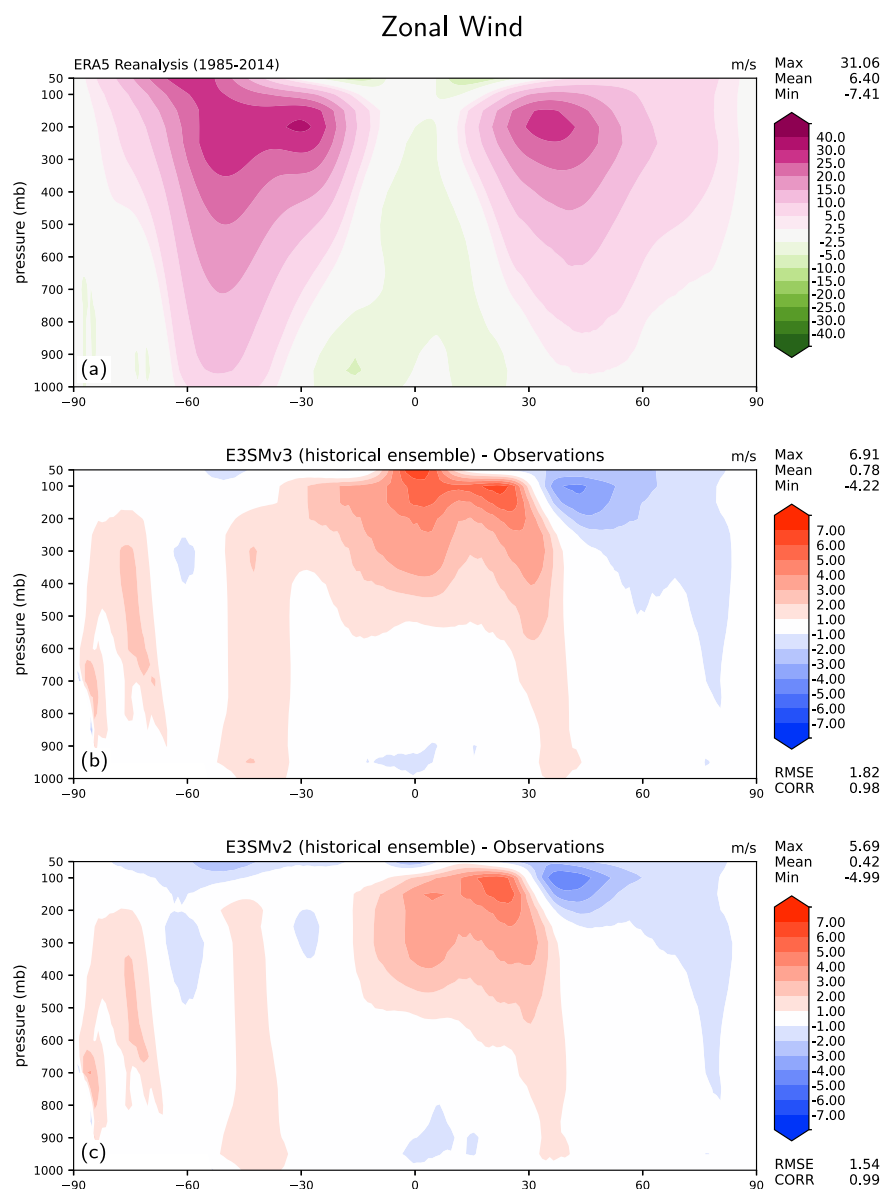


Figure 23. Vertical distribution of zonal mean wind of 1985–2014 climatology from (a) ERA5, and mean biases in panel (b) E3SMv3 and (c) E3SMv2.

differences emerge from coupled feedbacks attributable to the changes in other components and coupling fields. Key regions of ocean surface climate improvements are discussed below.

The improvement in the historical SST is due to a wide band of warming in the Northern Hemisphere (Figure 25d). This warming is partly due to the ocean changes from v2 to v2.1, but with the largest contribution from the atmospheric changes in v3. Specifically, the changes to the aerosol forcing in v3 produces a much more realistic surface air temperature evolution in the historical period (Figure 11b). This warmer climate, focused in the Northern Hemisphere mid-latitudes, reduces the cold SST bias in the North Atlantic and Pacific without deteriorating the Southern Hemisphere SST (which has a mean warm bias), resulting in a net improvement in the global patterns of SST.

It is also interesting to compare Figure 25d with Figure 29d in Golaz et al. (2022). That earlier figure presented a hypothetical reconstruction of E3SMv2 historical simulations by compositing *hist-GHG* and *hist-aer* using weights optimized to reduce the mismatch between the simulated and observed historical temperature record,

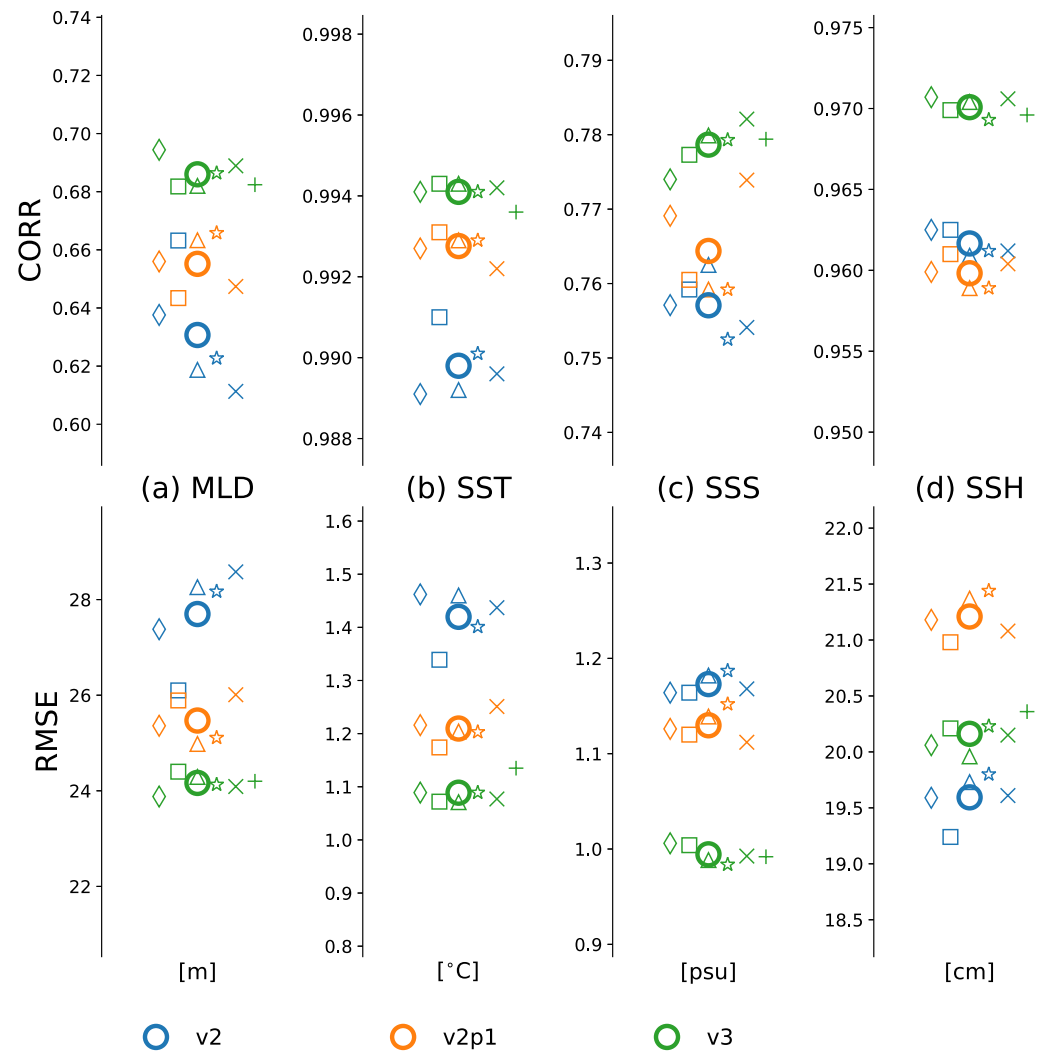


Figure 24. Spatial correlations (top row) and RMSE (bottom row) of the global ocean (a) MLD (m), (b) SST ($^{\circ}\text{C}$), (c) SSS (psu), and (d) SSH (cm) for the v3 (green markers), v2.1 (orange markers), and v2 (blue markers) configurations. Individual thin markers indicate the metric obtained from individual ensemble members and the bold open circle markers indicate the multi-realization averages of the five v3/v2.1/v2 historical ensemble members. For v3, an additional historical member (+) was run using a 15 min baroclinic time step. The model variables and observational data sets were regridded to a common $0.5^{\circ} \times 0.5^{\circ}$ regular grid. The observational data sets (Hadley, Aquarius, and Argo, respectively) are described in the Section 4.4 text.

effectively reducing the aerosol-related cooling. There is a strong similarity in the northern hemisphere warming pattern from the reduced aerosol-related cooling. But additionally, E3SMv3 also reduces warm biases in the southern hemisphere, leading to overall smaller mean bias (0.11 K vs. 0.20 K) and RMSE (1.06 K vs. 1.17 K).

Another region of stark improvement is the ocean state in the Mediterranean Sea. In E3SMv2, the salty bias in that region is associated with mixed layer depths much deeper than observed (Figures 26c–26e). This is consistent with a denser surface water, leading to convective mixing. In E3SMv3, the surface state is fresher and closer to observations, bringing the mixed layer depths to a more realistic range (Figures 26d–26f). Although model biases are best assessed against the observational time period, the changes in Mediterranean Sea surface salinity are already visible in the spin-up of the *piControl* simulations (Figure 27a). The major bias reduction in the Mediterranean Sea SSS occurs between v2.1 and v3, thus is not due to the additional ocean dynamics introduced in v2.1, but rather the v2.1–v3 coupled developments. The salinity difference is not present in the initial conditions but emerges early (<200 years) in the spin-up. In the first decade of spin-up, the improvements to the v3 land model (its initialization from a spun-up state, and its higher resolution) produce a stronger runoff in the

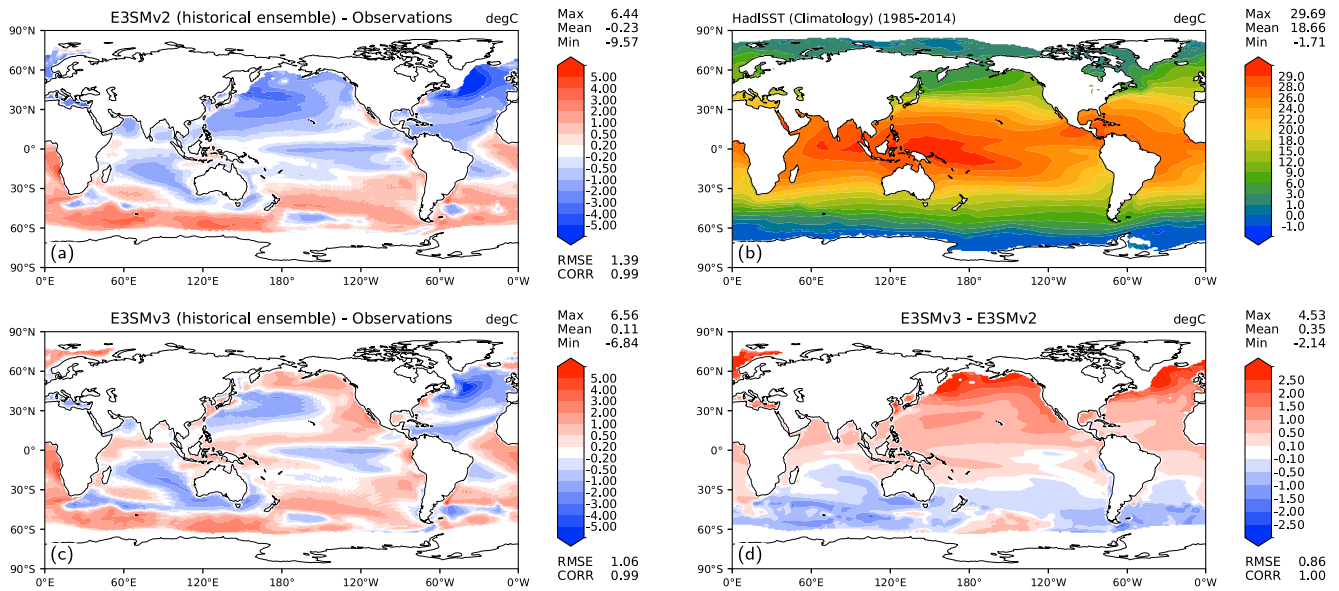


Figure 25. 1985–2014 annual mean sea surface temperature (SST) biases in (a) E3SMv2 relative to (b) HadISST climatology, and biases in panel (c) E3SMv3 and (d) the differences between E3SMv3 and E3SMv2.

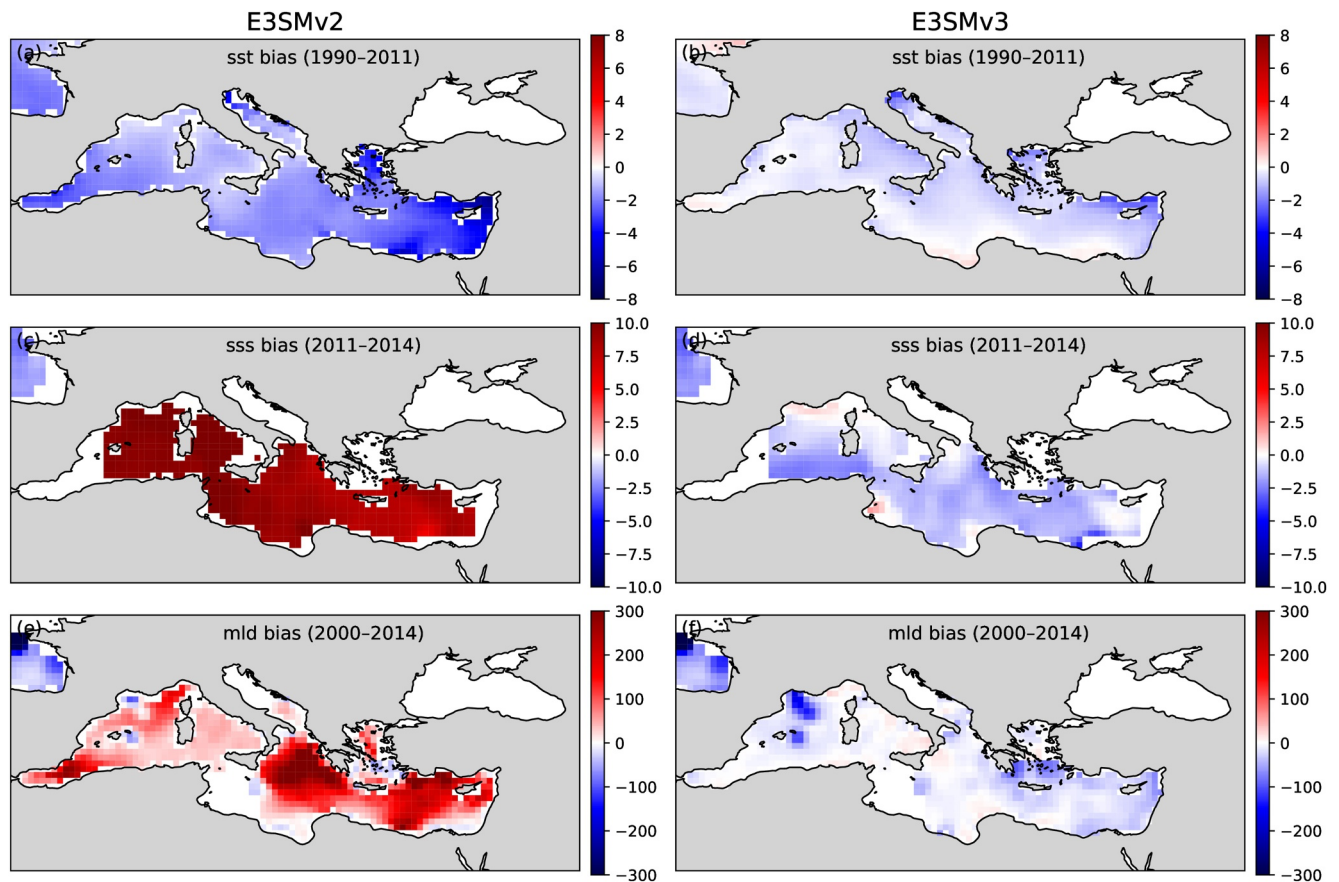


Figure 26. Spatial maps of the Mediterranean Sea historical biases in panels (a, b) sea surface temperature, (c, d) sea surface salinity and (e, f) mixed layer depth in E3SMv2 (left) and E3SMv3 (right). The model variables and observational data sets were regridded to a common $0.5^\circ \times 0.5^\circ$ regular grid. The observational data sets (Hadley, Aquarius, and Argo, respectively) are described in the Section 4.4 text.

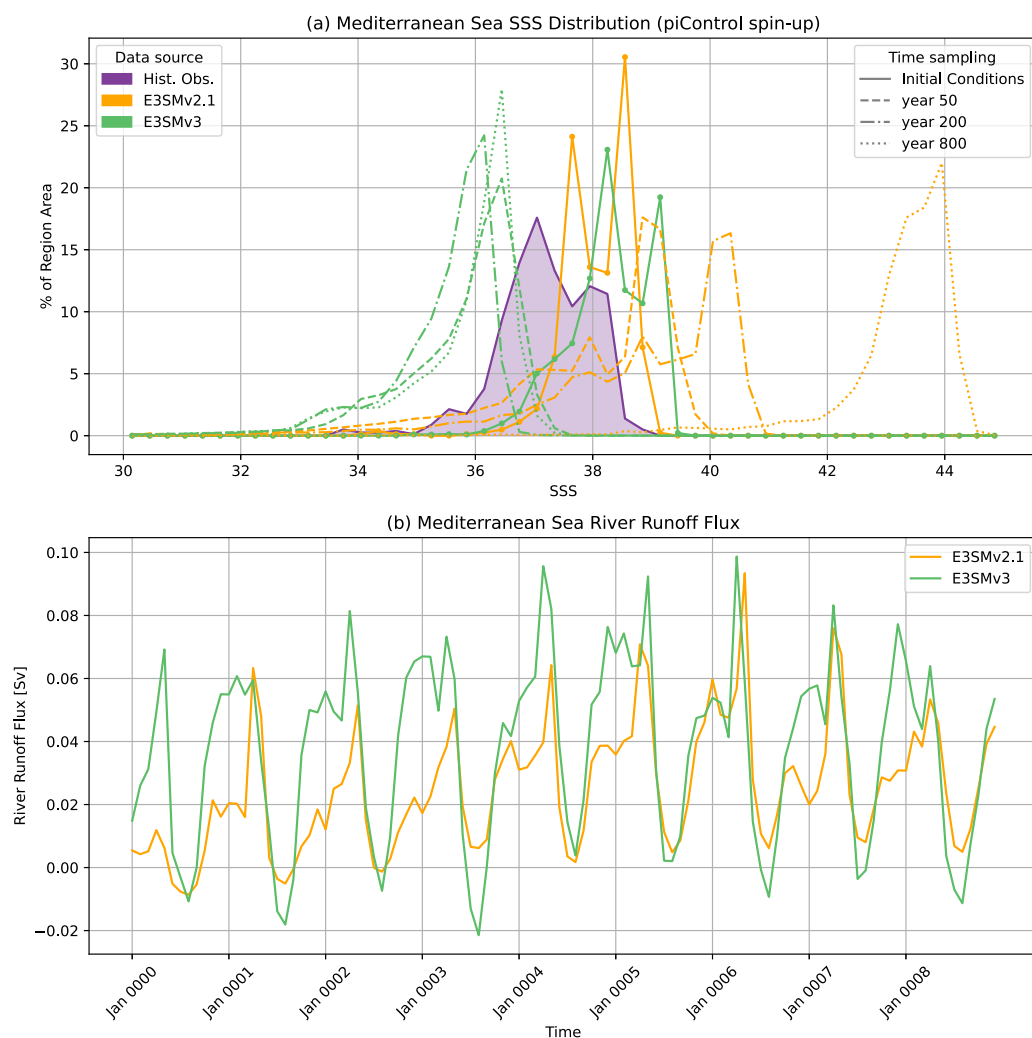


Figure 27. Mediterranean Sea evolution in the E3SMv2.1 (orange) and E3SMv3 (green) spin-up of the *piControl* simulations. (a) Distribution of the sea surface salinity from the initial conditions state (solid lines) to the 12-month distribution of years 50, 200, 800 (see legend). The range of the (regridded) historical SSS climatology used in Figure 26 is plotted for reference (purple). (b) River runoff flux into the Mediterranean Sea (in Sv) in the first 9 years of the simulation.

Mediterranean Sea (Figure 27b) which freshens the Mediterranean Sea, improving its fidelity in v3 compared to v2.1 or v2 (not shown), which develop a strong salty bias. Other freshwater fluxes (precipitation, evaporation) do not exhibit a significant difference in that period (not shown).

With the inclusion of prognostic vegetation growth, the land model simulates seasonal, interannual, and longer timescale variations in total leaf area index (TLAI) for each plant functional type (PFT) in each land grid cell. TLAJ is defined as the area of leaves per unit area of ground. In earlier model versions (E3SMv1 and E3SMv2), TLAJ was prescribed by a single monthly climatology for each PFT in each grid cell. Because that climatology was based on satellite remote sensing inputs, and because the remote sensing instruments and algorithms tend to give a saturated response for dense vegetation canopies, the TLAJ from earlier model versions is expected to be biased low compared to ground-based observations, with that bias especially evident over dense forest regions (Fang et al., 2019). ELMv3 predicts higher TLAJ over dense tropical forest regions, compared to the remote sensing driven diagnostic quantity for ELMv1 (Figure 28). One consequence of higher TLAJ for tropical forests in ELMv3 is that the energy balance over these regions is shifted toward more latent heat flux (Figure 28). While the increase in TLAJ over tropical forest in ELMv3 can be about a factor of two, increases in latent heat flux over those regions are more modest, around 25%. Other regions outside the tropics also show differences in TLAJ, both increases and decreases, and changes in latent heat flux which are usually but not always in the direction expected

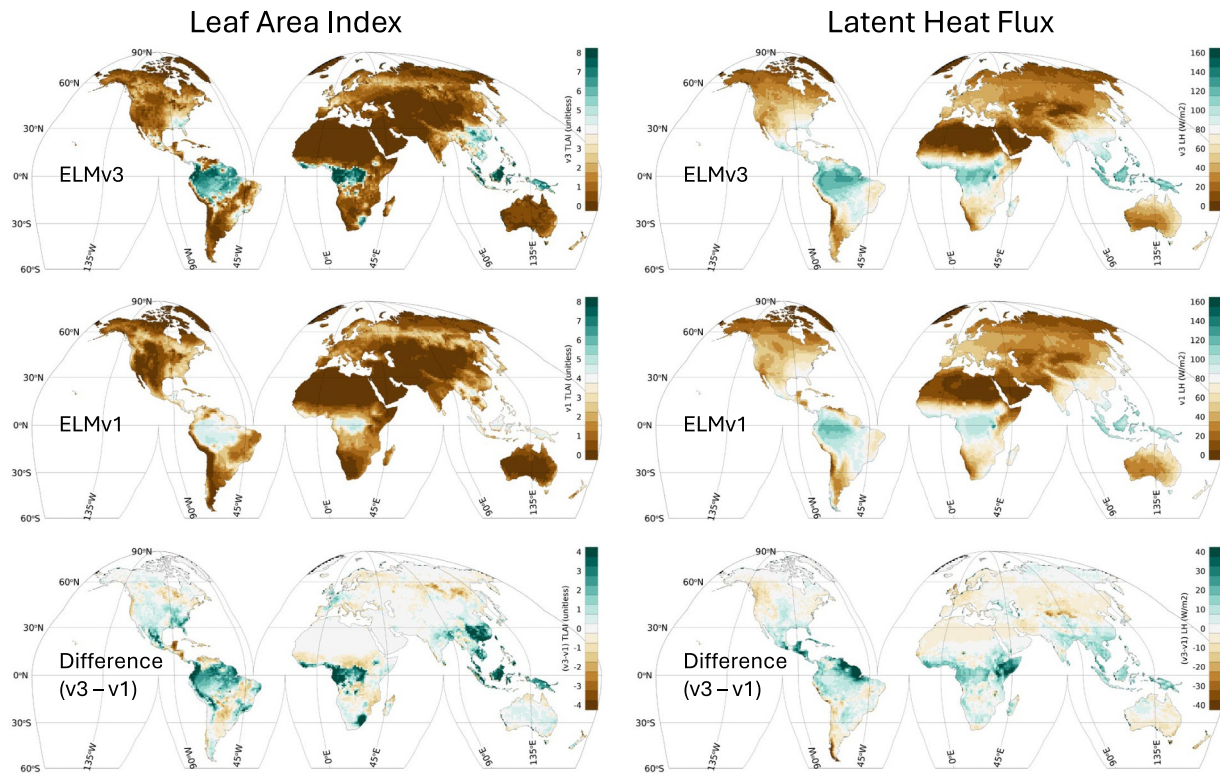


Figure 28. Total leaf area index (TLAI, left panels) and latent heat flux (or LH, right panels) shown for both variables as multi-year annual averages over the simulated period 1985–2014. Upper panels show the results from E3SMv3 (using ELMv3), middle panels show the results from E3SMv1 (using ELMv1), and lower panels show the difference between the two simulations (v3–v1).

from changes in vegetation. We are continuing to explore these regional variations, and we expect to find sensitivity to prescribed changes in human land use.

4.5. Tropical Variability

With recent updates in atmospheric physics, particularly enhancements to the ZM deep convection scheme, the representation of tropical variability in E3SMv3 has improved substantially compared to E3SMv2. To illustrate these advances, we evaluate the model's simulation of the Madden–Julian Oscillation (MJO, Madden & Julian, 1971, 1972), a primary mode of tropical intraseasonal variability characterized by eastward-propagating convection from the Indian Ocean to the western Pacific during boreal winter. In observations, cross-lag correlation analyses reveal pronounced eastward propagation of MJO signals from 60°E to 160°E, along with meridional propagation extending from the equator to approximately 20°S and 20°N (Figure 29a).

Cross-lag correlation of precipitation and 850 hPa zonal wind indicates that E3SMv2 exhibits substantial stagnation near 80°E (top panel of Figure 29b), failing to capture the dominant eastward propagation across the Indian Ocean basin observed in the ERA-Interim reanalysis. In particular, the simulated MJO in E3SMv2 weakens significantly around 120°E, suggesting difficulty in propagating across the Maritime Continent. Additionally, E3SMv2 simulates limited meridional propagation of convection (bottom panel of Figure 29b), further deviating from key observed characteristics of the MJO. In contrast, E3SMv3 captures both the eastward and meridional propagation of the MJO (Figure 29c) much more realistically, representing a marked improvement over E3SMv2.

The spectral characteristics of the MJO are evident in outgoing longwave radiation (OLR) as eastward-propagating signals in the tropics, with periods between 30 and 90 days and zonal wavenumbers between 1 and 3 during the boreal winter (Figure 30a, NOAA Interpolated Outgoing Longwave Radiation, Liebmann and Smith (1996)). While E3SMv2 exhibits substantial spectral power in eastward-propagating modes with periods longer than 90 days, it shows little power in the key MJO band of 30–90 days (Figure 30b), indicating a deficiency in simulating the dominant MJO timescales. In contrast, E3SMv3 greatly enhances spectral power within this

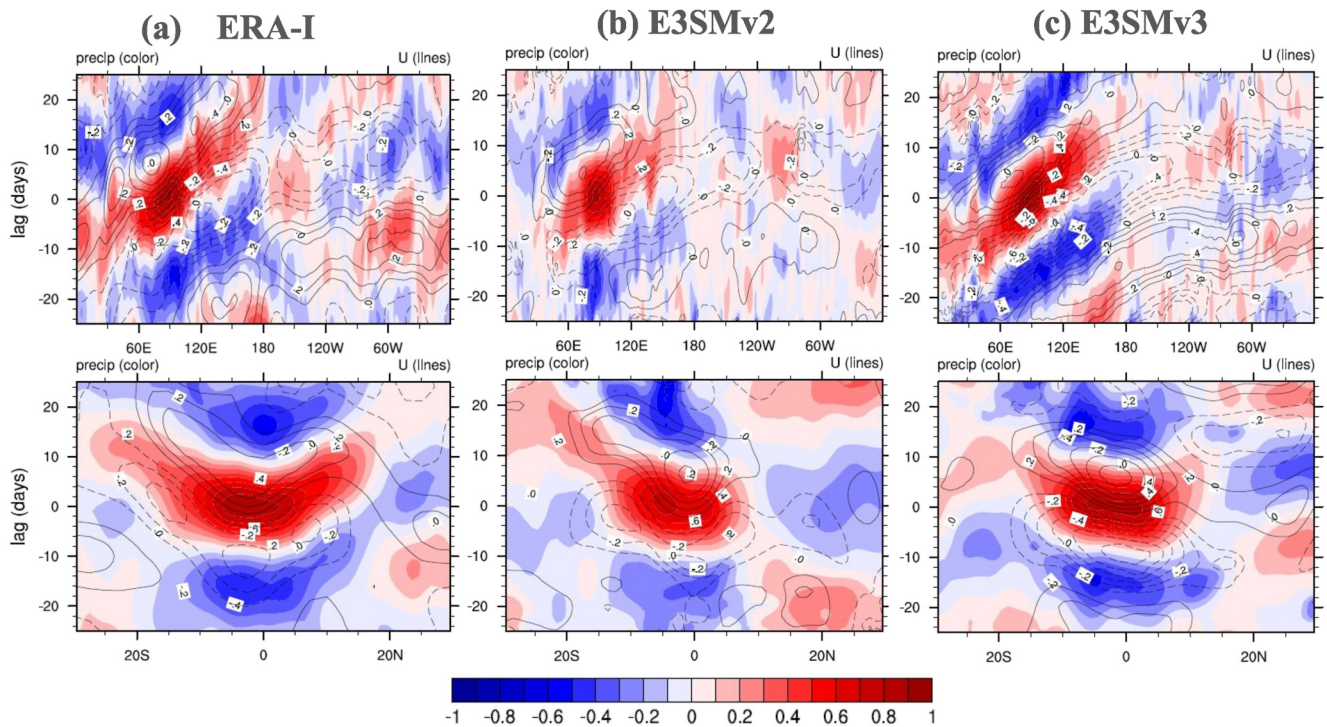


Figure 29. Cross-lag correlation of precipitation and zonal wind at 850 hPa during the boreal winter by (a) ERA-Interim, and five-member ensemble means of (b) E3SMv2 and (c) E3SMv3. The cross-lag correlation is computed using a time series index of 20–100-day Lanczos bandpass-filtered precipitation within domain (10°S–10°N, 80°E–100°E). The latitude averaging range for the top panels is 10°S and 10°N, and the longitude averaging range for the bottom panels is 80°E and 100°E.

critical 30-to-90-day band (Figure 30c), reflecting a significant improvement in representing the MJO's temporal variability. Recent work on MJO diversity (e.g., Back et al., 2024) suggests that differences in the relative frequency of MJO types can affect spectral characteristics. In particular, a higher proportion of stagnant or near-stagnant MJO disturbances tends to project power onto periods longer than 90 days. The increased peak in E3SMv3 compared with E3SMv2 could therefore be related to changes in the distribution of MJO types. A detailed analysis of MJO diversity in E3SMv3 is beyond the scope of this study but will be an important next step to evaluate this hypothesis.

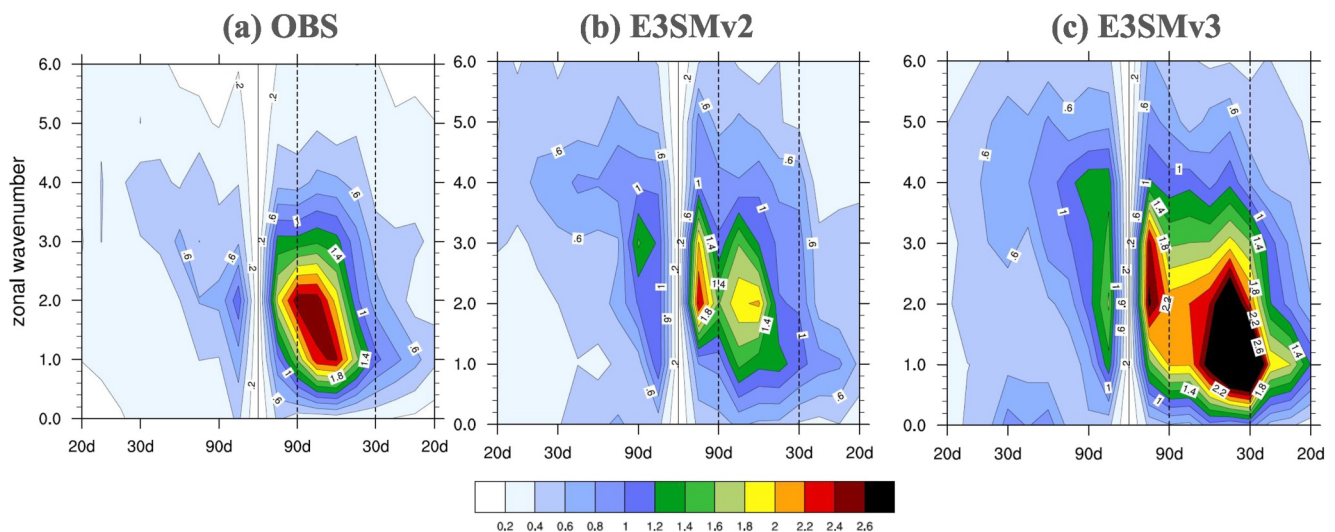


Figure 30. Wave spectra of outgoing longwave radiation (OLR): (a) observation, (b) five-member ensemble mean of E3SMv2, and (c) five-member ensemble mean of E3SMv3.

5. Conclusion

E3SMv3 incorporates extensive updates across its components to improve model fidelity and address long-standing biases. This third major release builds upon the foundation laid by its predecessors, introducing innovations in atmospheric physics, sea ice modeling, land processes, and coupling mechanisms. Collectively, these updates enhance the model's ability to simulate climate trends with greater accuracy.

The atmospheric component underwent substantial improvements, including updates to chemistry, aerosol-cloud interactions, convection scheme, and microphysics. The land model now includes dynamic vegetation growth and a new representation of sub-grid topographic effects on incoming solar radiation. The ocean component adopted a new time-stepping scheme but remains otherwise unchanged from the intermediate release of E3SMv2.1. The sea ice model features advanced snow physics and additional capabilities, such as wind redistribution of snow, to improve cryospheric simulations.

The ocean and sea ice grid now features increased resolution, with cell sizes ranging from 20 to 30 km, nearly doubling the number of grid cells compared to E3SMv2 (Figures 1 and 2 and Table 1). An innovation in E3SMv3 is the adoption of a tri-grid configuration where the land and river components share a common 0.5° latitude-longitude grid—a significant departure from previous versions. This change increases the number of land grid cells by over an order of magnitude. To minimize grid imprinting and improve coupling between the atmosphere and surface processes, a new non-linear remapping algorithm was developed (Figure 3). Overall, E3SMv3 is approximately 50% computationally more expensive than its predecessor E3SMv2 (Figure 4).

The E3SMv3 pre-industrial control simulation was initialized following a 2,000-year spin-up to achieve quasi-equilibrium. Over its 500-year duration, the control simulation exhibits minimal change in total OHC (Figure 5a) and regional SST patterns remain stable over time (Figure 7). The correction of a surface pressure coupling error in E3SMv3 led to the removal of unrealistic SSH features near continental coasts (Figure 8).

Transient climate response (2.33 K) and effective equilibrium climate sensitivity (3.90 K) in E3SMv3 are comparable to v2 and remain on the higher side relative to CMIP6 models (Figure 10).

E3SMv3 historical simulations demonstrate marked improvements in reproducing observed temporal trends, particularly addressing biases present in earlier versions. Reducing the aerosol-related cooling (Figures 12 and 13) was the top priority and it improved the model's ability to simulate historical surface temperature record, resolving the “pothole cooling” bias that plagued previous versions (Figure 11). OHC trends now closely align with observational records (Figure 14), and sea ice evolution exhibits more realistic behavior (Figure 15). These improvements highlight the model's enhanced ability to capture the complex interactions between anthropogenic forcing and Earth system components.

Despite these advancements, the evaluation of present-day climate reveals both strengths and weaknesses. E3SMv3 performs competitively in reproducing global climatology for surface radiative fluxes, precipitation, and air temperature, with errors comparable to or smaller than those of CMIP6 multi-model means. However, biases persist in certain dynamical fields, such as sea level pressure and upper-level winds (Figures 17 and 18). Top-of-atmosphere radiative fluxes (Figures 19 and 20) remain similar between E3SMv2 and v3 with modest improvements in outgoing longwave radiation. Structural biases in precipitation (Figure 21) are also largely similar between E3SMv2 and v3, but with noticeable improvements over the continental United States and tropical land masses and along North Atlantic storm track. Additionally, improvements in zonal mean temperature are evident (Figure 22).

Observed mean ocean surface and near-surface climate is better represented in E3SMv3 compared to earlier versions as evidenced by higher correlations and lower errors for MLD, SST and SSS (Figure 24), though SSH shows a slight degradation. Regional SST patterns (Figure 25) exhibit significant reduction in cold biases in the North Pacific and Atlantic, a result of the reduction in the excessive aerosol-related cooling seen in previous versions. The inclusion of dynamic vegetation growth has a significant impact on TLAI compared to the prescribed seasonal cycle in earlier versions. TLAI increases in tropical regions, accompanied by a corresponding rise in surface latent heat flux (Figure 28).

A development priority was to improve tropical variability compared to E3SMv2. Improvements are illustrated in Part 1 and here with more realistic representation of the MJO cross-lag correlation and spectrum (Figures 29 and 30).

Overall, E3SMv3 represents a significant step forward, offering a more credible tool for studying climate and interactions across multiple components. The five-member historical ensemble analyzed here has been augmented to a 25-member ensemble, which will be the focus of an upcoming Part 3 study, particularly with respect to variability. Furthermore, the present standard-resolution configuration of E3SMv3 will be augmented with regionally refined and higher-resolution configurations. This third generation of E3SM model configurations will be the last for the legacy Fortran code base. The fourth generation of E3SM models will transition to new atmosphere and ocean components written in C++/Kokkos, enabling performance portability across CPU and GPU computer architectures.

Appendix A: Nonlinear Maps

The key ideas of the nonlinear map algorithm are as follows. First, we describe the two linear maps the method uses. Let A_L be the matrix for the conventional first-order, finite-volume map and A_H be the matrix for the second-order, finite-volume map. Because the two methods are closely related, the nonzero pattern of A_L is inside that of A_H ; that is, if $A_L(i, j) \neq 0$, then $A_H(i, j) \neq 0$ as well. Thus, A_H provides data for every target cell that A_L does; and, if a source cell contributes to a target cell using A_L , then it does when using A_H , as well.

Second, we need a means to determine what the true global mass is on the target grid. Let a_T be the vector of target cell areas, and let x be a vector of fluxes on the source grid. The product of A_L and x , $y_L \equiv A_L x$, gives the low-order flux on the target grid. Then $M \equiv a_T \cdot y_L$, the dot product of a_T and y_L , is the global integral of the flux on the target grid. This is the true mass because A_L is conservative.

Third, we need to define *shape preservation*. Let u be a vector of upper bounds. $u(i)$ is the largest value permitted in target cell i . $u(i)$ is calculated as

$$u(i) \equiv \max_{j \in [j | A_H(i, j) \neq 0]} x(j).$$

That is, the target value cannot be larger than the contributing source values. l is defined and computed similarly for lower bounds. In addition, $l(i)$ and $u(i)$ are set to zero if $y_L(i) = 0$; thus, the high-order flux is zero wherever the low-order flux is.

Finally, with these definitions and data, we can then call CLIPANDASSURED SUM, described in Bradley et al. (2019). Let $\text{diag}(a_T)$ be a diagonal matrix with a_T on its diagonal. CLIPANDASSURED SUM uses one global reduction to solve the following minimization problem:

$$\begin{aligned} y &\equiv \arg \min_y && \|\text{diag}(a_T)(y^* - A_H x)\|_1 \\ \text{subject to} &&& a_T \cdot y^* = M \\ &&& l \leq y^* \leq u, \end{aligned}$$

where the inequalities apply element-wise in the final line. That is, y satisfies the conservation and shape preservation constraints by deviating from the high-order field $y_H \equiv A_H x$ by a mass redistribution whose magnitude is as small as possible.

In the standard-resolution configuration described in this article, linear maps are used for precipitation fluxes rather than nonlinear maps for the following reason. Each flux map is conservative in isolation. However, regardless of map type, the surface coupler did not originally conserve atmosphere-to-surface fluxes overall in the case of multiple surface grids having partial cell overlaps. The coupler maintains budgets for water, heat, and other quantities that quantify the nonconservation. It was found that nonlinear maps increased this already-present nonconservation in precipitation fluxes to an unacceptable level. This problem appeared quantitatively in the coupler's water budget. The heat budget was acceptable; thus, nonlinear maps are used to map radiation fluxes. Subsequent work for the high-resolution model configurations addressed this issue in the surface coupler; thus, in the high-resolution configurations, nonlinear maps are used to map all fluxes. However, that additional algorithm was not available in the standard-resolution model configuration and simulations described in this article.

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The algorithm to address the overall atmosphere-to-surface nonconservation leverages the nonlinear-map capability; that is, overall atmosphere-to-surface conservation relies on the use of nonlinear maps, whether with high-order or low-order linear maps. (Overall conservation is possible with linear maps alone, but such an approach within the framework of cpl7/MCT will cause unacceptable artifacts at coastlines.) The procedure is as follows. First, partition the source field on the atmosphere grid between land and ocean/sea ice grids. Second, require each of these two surface grids to accept exactly that grid's partition of source mass. These two steps are implemented by modifying the definitions of a_T and M . Third, set $l(i)$ and $u(i)$ to zero wherever the surface grid's fraction is zero. This procedure uses internally consistent grid-cell fractions for land and ocean/sea ice on these and the atmosphere grids, all derived from the ocean/sea ice grid-cell fractions.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

All model codes may be accessed on the GitHub repository at <https://github.com/E3SM-Project/E3SM>. A maintenance branch (maint-3.0; <https://github.com/E3SM-Project/E3SM/tree/maint-3.0>) has been specifically created to reproduce these simulations. Bit-for-bit results with the original simulations on identical machines will be maintained on that branch for as long as the computing environment supports it. E3SMv3 data access is documented at https://docs.e3sm.org/e3sm_data_docs/_build/html/v3/CoupledSystem. Complete native model output has been archived and is accessible directly on NERSC at <https://portal.nersc.gov/archive/home/projects/e3sm/www/CoupledSystem/E3SMv3/LR>. A subset of the data reformatted following CMIP conventions also available through ESGF at <https://metagrid.esgf-west.org/search?project=CMIP6-E3SM-Ext>.

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language editing of this article. This paper describes objective technical results and analysis. Any subjective views or opinions that might be expressed in the paper do not necessarily represent the views of the U.S. Department of Energy or the United States Government.

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Erratum

The originally published version of this article contained a typographical error. Coauthor Johannes Muelmenstaedt's name was incorrectly spelled Johannes Mülmenstädt. The error has been corrected, and this may be considered the authoritative version of record.